Brain Interacts

BRAIN INTERACTS RESEARCH · MONOGRAPH

The Behavioral Enterprise

Trait Interaction, the Big Five, and Workforce Intelligence

LEAD AUTHOR

Srishti Sharma

CO-AUTHOR

Sandeep Kumar

PUBLISHED BY

Brain Interacts

BEFORE YOU BEGIN

About this document

This monograph is a research document prepared by Brain Interacts for readers of the Brain Interacts website. It sets out the science behind THE BRAIN, our Behavioural Reasoning and Interaction Network, and behind Corporate Brain, the workforce intelligence platform that puts that science to work.

It is not a peer-reviewed publication. It is an open account of how we think, what we have built, what the evidence supports, and what we are still testing.

We have written it for organizational psychologists, people-analytics leaders, HR practitioners, business leaders, researchers in organizational behaviour, and graduate students. It assumes basic familiarity with personality assessment but no statistical training beyond the introductory level. Where a statistical idea matters, we explain it in plain language.

THREE CONVENTIONS

ILLUSTRATIVE COMPOSITE
Worked examples marked this way explain the framework but do not describe real people or real organizations.

RESEARCH IN PROGRESS
Panels marked this way show where evidence is being collected and what it will need to show. We have not filled those gaps with invented numbers.

NO CLIENT NAMED
No client organization is named. Industries are described, never companies.

The document publishes the taxonomy of trait interactions, the architecture of THE BRAIN, the domain rules in outline, and the full catalogue of 58 expert-defined interaction pairs. The detailed engine instructions, the full Tier 2 rule weights and the complete item bank remain proprietary to Brain Interacts.

HOW TO READ THIS DOCUMENT

Part I	Explains why the Big Five is the right foundation.
Part II	Presents THE BRAIN.
Part III	Moves from individuals to organizations.
Part IV	Covers validity, ethics and future work.
Part V	A full review of the research base.
Appendices	The trait inventory, the flags, the modifiers and the complete Tier 1 interaction catalogue.

PREFACE

Why we wrote this

This work began with a frustration that most practitioners in people assessment will recognize. The scores were good. The instruments were valid. The reports were tidy. Yet the conversations that followed were often unsatisfying. A manager would read that an employee scored 8 on Ambition and nod. When that employee later derailed under pressure, the report had nothing to say about why.

The answer, we came to believe, was not in any single score. It was in what the scores did to one another. Ambition reads one way beside strong Self-Regulation and another way beside weak Self-Regulation. Agreeableness reads one way beside strong Accountability and another way when it quietly overrides it. Psychologists have known for nearly a century that traits interact. What the field lacked was a systematic, shared way to describe those interactions and to apply them at the scale modern organizations require.

THE BRAIN is our attempt to supply that. This document sets out its theory, its architecture and its use inside a working platform, Corporate Brain. It also sets out its limits. We have tried throughout to separate what is established, what is proposed, and what still needs evidence. We think that separation is the most useful thing a research document from a technology company can offer. Readers have every right to be sceptical of vendors who describe their own products. The best answer to that scepticism is to show the reasoning, show the evidence, and show the gaps.

We owe a great deal to the researchers whose work we draw on, and to the organizations and employees who have taken part in early deployments. Any errors of interpretation are ours.

Srishti Sharma and Sandeep Kumar

ABOUT THE AUTHORS

Srishti Sharma

LEAD AUTHOR

The lead author of this monograph and the principal architect of THE BRAIN framework at Brain Interacts. Her work focuses on personality structure, trait interaction and the translation of psychometric evidence into practical guidance for managers and employees. She led the development of the Trait DNA layer, the eleven-type interaction taxonomy and the expert specification of the 58 Tier 1 interaction pairs.

Sandeep Kumar

CO-AUTHOR

Vice President of Sales and Marketing at Brain Interacts. He leads the commercial development of Corporate Brain and works directly with client organizations across automotive, banking, insurance, broking, IT services and urban mobility. His contribution centres on the link between behavioural intelligence and business strategy: the goal-first design of Corporate Brain, the goal-by-function mapping and the strategy library described in Part III.

CONTENTS

What this monograph covers

Part I · Foundations

Chapter 1 Introduction: Measuring Traits Is Not Understanding Behaviour
Chapter 2 The Big Five: Origins, Structure and Evidence
Chapter 3 Competing Models and the Limits of Single Traits

Part II · The BRAIN Framework

Chapter 4 What the Big Five Predicts at Work
Chapter 5 A Taxonomy of Trait Interactions
Chapter 6 The BRAIN Architecture: Trait DNA and the Interaction Matrix
Chapter 7 The Narrative Engine: A Constrained Reasoning Instrument

Part III · From Individuals to Organizations

Chapter 8 Archetypes: From Profiles to Patterns
Chapter 9 Corporate Brain: Linking Behaviour to Business Strategy

Part IV · Responsibility and the Future

Chapter 10 Validity, Ethics and Governance
Chapter 11 Future Directions and Conclusion

Part V · Evidence Review

Section A Foundations of the Big Five
Section B Persons, Situations and States
Section C The Interaction Question
Section D Outcomes Beyond the Job
Section E Digital Traces and Measurement
Section F How People Read and Use Reports
Section G Teams as Cognitive Systems
Section H Analysis of THE BRAIN's Own Materials
Section I Synthesis

Appendices

A Trait Inventory by Domain
B Tier 1 Catalogue Summary
C Non-negotiable Flags
D Modifiers and Facet Metrics
E Research Programme
F The Complete Interaction Matrix
Glossary

ABBREVIATIONS

BRAIN	Behavioural Reasoning and Interaction Network	**GDPR**	General Data Protection Regulation (EU)
CAPS	Cognitive-Affective Processing System	**HRMS**	Human Resource Management System
CFA	Confirmatory factor analysis	**KPI**	Key performance indicator
DPDP	Digital Personal Data Protection Act, 2023 (India)	**NEO**	NEO Personality Inventory (Costa and McCrae)
FFM	Five-Factor Model	**STEN**	Standard ten score scale (1 to 10)

PART I

Foundations

Chapter 1	Introduction: Measuring Traits Is Not Understanding Behaviour
Chapter 2	The Big Five: Origins, Structure and Evidence
Chapter 3	Competing Models and the Limits of Single Traits

CHAPTER 1

Introduction: Measuring Traits Is Not Understanding Behaviour

SUMMARY Personality assessment has become very good at measuring traits and much less good at explaining behaviour. This chapter sets out the gap between the two, argues that the gap lies largely in trait interaction, explains why the gap grows as assessment scales across organizations, and introduces the four contributions of this monograph.

The organizational cost of misreading people

Organizations make thousands of people decisions every year. They decide whom to hire, whom to promote, who leads which team, who receives coaching, and who is quietly at risk of leaving or burning out. Each decision rests on a reading of a person. When the reading is wrong, the costs are concrete: a failed senior hire, a high performer who derails at the moment of greatest responsibility, a team that never gels, a capable employee who exhausts themselves without anyone noticing.

The costs are rarely confined to the person involved. A senior hire who fails takes months to identify, months more to replace, and often leaves a team weakened by lost trust. A high performer who derails under pressure can damage client relationships built over years. An employee who quietly burns out takes with them knowledge that never made it into any system. These are not abstract losses. Most leaders can name several from their own experience, and most of those stories share a feature: the warning signs were present in hindsight but not legible at the time.

Psychometric assessment was built to make those readings more accurate. Over the past century it has largely succeeded at its first task, which is measuring individual differences reliably. Modern Big Five instruments produce scores that are stable over time, consistent across languages, and meaningfully related to performance, leadership and wellbeing (Barrick and Mount, 1991; John, Naumann and Soto, 2008).

Yet anyone who has sat with a manager and a personality report knows the next step is harder. A score is not a decision. The manager wants to know what this person will actually do: under pressure, in a new team, when targets slip, when a colleague pushes back. Answering that requires reasoning about how the person's traits work together. Most reports leave that reasoning to the reader.

Same score, different person

Consider two employees who both score 8 out of 10 on Ambition.

ILLUSTRATIVE COMPOSITE

Employee A · Ambition 8, Self-Regulation 3. In calm periods A is energetic and productive. Under pressure, A's drive outruns their self-control: decisions become reactive, relationships suffer, and judgment slips in exactly the moments that matter most.

Employee B · Ambition 8, Integrity 8. B pursues goals relentlessly but refuses to compromise standards to reach them. Most organizations would value this profile very highly.

The Ambition score is identical. The people are not. A report that reads each trait separately describes A and B in almost the same words. The difference between them lives entirely in the interaction.

The same logic applies to low scores. A low score on Ambition might describe a disengaged employee, or it might describe a disciplined, reliable contributor whose strong Self-Regulation turns modest drive into steady delivery within a defined scope. A report that flags low Ambition as a gap, without checking what sits beside it, can do real harm to a dependable person's prospects. Reading scores in isolation does not only miss risks. It also misses strengths.

Why the problem grows with scale

When an experienced psychologist assesses a few people they know well, interaction reasoning happens informally. The practitioner holds the whole profile in mind and draws on years of pattern recognition. This does not scale. When an organization assesses hundreds or thousands of employees, reports are read by managers with no psychological training. Even trained practitioners reading the same profile can reach different conclusions, because no shared vocabulary exists for how traits combine.

The result is a paradox. As assessment becomes more widespread, the depth of interpretation tends to fall. Organizations gain data and lose understanding. A dashboard of five domain scores for every employee feels like knowledge, but it rarely changes a decision, because the decision depends on what the scores mean together, and the dashboard does not say.

There is a second scaling problem. Consistency matters for fairness. If two employees with the same profile receive different interpretations depending on which practitioner reads their report, the assessment process is not treating them equally. A systematic way of reasoning about interactions is therefore not only a matter of insight. It is also a matter of fairness and auditability.

The argument of this monograph

This monograph makes four claims.

The Big Five is the right foundation	Among available models it has the strongest evidence for reliability, cross-cultural replication and workplace prediction (Chapters 2 to 4).
Single traits are not enough	Behaviour emerges in part from how traits interact. Those interactions can be classified into eleven distinct types, each with its own mechanism and behavioural signature (Chapter 5).
Interaction reasoning can be systematized	A tiered architecture, THE BRAIN, makes interaction analysis consistent and traceable across a 91-trait inventory (Chapter 6). A tightly constrained reasoning engine, rather than a free-form writer, can apply that architecture at scale (Chapter 7).
Individual insight must connect to organizational goals	Archetypes and strategy mapping translate profiles into decisions about work allocation, management and communication (Chapters 8 and 9).

The monograph is also explicit about limits. The second claim is interpretive and theoretical. Its predictive value is a hypothesis, and Chapter 3 and Part V explain why the strongest available evidence makes that hypothesis demanding. Chapter 10 sets out the validation evidence still required and the ethical safeguards any such system needs.

Research questions

RQ1	What justifies the Big Five as the foundation for organizational behavioural analytics compared with competing models?
RQ2	Can trait interactions be classified into a finite, theoretically grounded set of types?
RQ3	Can a tiered architecture apply interaction reasoning consistently across a large trait inventory?
RQ4	Under what constraints can an automated reasoning engine apply such an architecture reliably and transparently?
RQ5	How can individual behavioural profiles be linked to organizational goals and functions?

What this monograph is not

It is worth being clear about three things this document does not attempt. It is not a manual for administering Corporate Brain; product documentation serves that purpose. It is not a clinical text; nothing in THE BRAIN is a diagnosis, and wellbeing indicators are treated throughout as prompts for support, not as findings about health. And it is not a claim that personality explains everything. Situations, skills, resources, leadership and luck all shape what people do at work. Personality is one important lens among several, and the value of a better lens depends on using it alongside the others.

How this monograph is organized

Part I establishes the foundations: the Big Five, its competitors and the long tradition of interactionist thinking. Part II presents THE BRAIN: the evidence on what the Big Five predicts, the interaction taxonomy, the architecture and the narrative engine. Part III moves from individuals to organizations through archetypes and the Corporate Brain platform. Part IV addresses validity, ethics and future directions. Part V is a source-by-source review of the research that underpins the framework, including a critical analysis of our own materials. The appendices contain the reference material: trait inventory, flags, modifiers, data collection plan and the complete Tier 1 interaction catalogue.

REFERENCES

Barrick, M. R., & Mount, M. K. (1991). The Big Five personality dimensions and job performance: A meta-analysis. *Personnel Psychology*, 44(1), 1–26.

John, O. P., Naumann, L. P., & Soto, C. J. (2008). Paradigm shift to the integrative Big Five trait taxonomy. In O. P. John, R. W. Robins & L. A. Pervin (Eds.), *Handbook of Personality* (3rd ed., pp. 114–158). Guilford Press.

CHAPTER 2

The Big Five: Origins, Structure and Evidence

SUMMARY This chapter traces how the Five-Factor Model emerged from the study of personality language, describes its five domains and thirty facets, reviews evidence on stability and change, and sets out why it is the most defensible foundation for organizational assessment.

From language to dimensions

The Big Five began with a simple idea, the lexical hypothesis. If a personality difference matters to people, language will develop words for it. The more important the difference, the more words it will attract, and the more languages will contain them. Galton first explored this in the 1880s. Allport and Odbert (1936) extracted around 18,000 personality-related terms from an English dictionary. Cattell reduced these to a smaller set of clusters, and later researchers, using factor analysis on ratings of these words, repeatedly found five broad dimensions (Tupes and Christal, 1961; Norman, 1963; Goldberg, 1990).

1880s	~18,000	Clusters	5
Galton explores the lexical idea	terms extracted by Allport and Odbert	Cattell reduces the list	broad dimensions recur in factor analysis

Factor analysis deserves a brief plain-language explanation, because it underlies almost everything in this field. When people rate themselves or others on many descriptive words, some words tend to be rated similarly: someone rated "talkative" is usually also rated "sociable" and "outgoing". Factor analysis is a statistical method that finds these clusters and summarizes them as a smaller number of underlying dimensions. The striking finding of the lexical tradition was that, across many samples and many word lists, five dimensions kept reappearing.

In parallel, Costa and McCrae developed questionnaire measures that converged on the same five domains and organized each into six facets (Costa and McCrae, 1992). The agreement between the lexical and questionnaire traditions, built by different researchers with different methods, is one of the main reasons the model gained acceptance.

The five domains and thirty facets

Table 2.1 summarizes the facet structure used in THE BRAIN's Big Five Trait DNA. Throughout this monograph, the fifth domain is scored in the direction of Emotional Stability (the reverse of Neuroticism), so that a high score is always the adaptive pole in organizational language.

DOMAIN	FACETS
O **Openness**	O1 Fantasy, O2 Aesthetics, O3 Feelings, O4 Actions, O5 Ideas, O6 Values
C **Conscientiousness**	C1 Competence, C2 Order, C3 Dutifulness, C4 Achievement Striving, C5 Self-Discipline, C6 Deliberation
E **Extraversion**	E1 Warmth, E2 Gregariousness, E3 Assertiveness, E4 Activity, E5 Excitement Seeking, E6 Positive Emotions
A **Agreeableness**	A1 Trust, A2 Straightforwardness, A3 Altruism, A4 Compliance, A5 Modesty, A6 Tender-Mindedness
N **Neuroticism, reported as Emotional Stability**	N1 Anxiety, N2 Angry Hostility, N3 Depression, N4 Self-Consciousness, N5 Impulsiveness, N6 Vulnerability

Table 2.1 Domains and facets (after Costa and McCrae, 1992).

Each facet in the Trait DNA carries a high-score description (STEN 7 to 10) and a low-score description (STEN 1 to 4), written as observable behaviour rather than definition. For example, high O1 Fantasy is described as a rich inner world with creative visualization, while low O1 is described as practical, literal thinking that prefers facts to imagination.

Openness

Openness describes receptiveness to new ideas, experiences, feelings and values. People high in Openness are curious, imaginative and comfortable with complexity. They enjoy abstract discussion, notice aesthetic qualities, and are willing to question established ways of working. People low in Openness prefer the familiar and the concrete. They value proven methods, practical outcomes and clear conventions. At work, high Openness supports innovation, learning and strategic thinking. Low Openness supports consistency, reliability in routine processes and a healthy scepticism of change for its own sake. Neither pole is better in general; the value depends on what the role needs.

Conscientiousness

Conscientiousness describes organization, dependability, persistence and the drive to achieve. High scorers plan, follow through and hold themselves to standards. Low scorers are more spontaneous and flexible, more comfortable with loose structure, and sometimes less reliable on detail. Conscientiousness is the most consistent single predictor of job performance across occupations, as Chapter 4 shows. Its facets differ in useful ways. Achievement Striving captures ambition and drive; Deliberation captures the tendency to think before acting; Order captures preference for tidiness and structure. Two people with the same Conscientiousness score can differ sharply on these facets, which is one reason THE BRAIN works at facet level where it can.

Extraversion

Extraversion describes sociability, assertiveness, energy and positive emotion. High scorers seek company, speak up, take the lead in groups and draw energy from interaction. Low scorers, often called introverts, prefer quieter settings, fewer and deeper relationships, and time to think before speaking. Extraversion predicts performance in sales and management roles, and it is the strongest correlate of leadership emergence. It is also the trait most often mistaken for competence. A confident speaker is not necessarily a good decision-maker, and Chapter 5 shows how the interaction of Extraversion-related traits with Integrity and Empathy can turn a strength into a risk.

Agreeableness

Agreeableness describes cooperation, trust, altruism and concern for others. High scorers are warm, helpful and conflict-averse. Low scorers are more competitive, sceptical and direct. High Agreeableness supports teamwork, service and conflict resolution. Very high Agreeableness can also suppress necessary challenge: a highly agreeable manager may avoid difficult conversations, accept poor standards, or fail to defend their team's interests. Low Agreeableness can support tough negotiation and independent judgment, but can damage relationships if it is not balanced by other traits.

Emotional Stability

Emotional Stability, the reverse of Neuroticism, describes calmness, resilience and freedom from persistent negative emotion. High scorers stay composed under pressure and recover quickly from setbacks. Low scorers experience more anxiety, irritability and vulnerability to stress. Emotional Stability predicts performance modestly across most jobs and is closely tied to wellbeing. It is also the domain that requires the most care in organizational use. Some of its facets, especially N3 Depression, touch on health. THE BRAIN treats those facets as indicators for confidential support, never as inputs to hiring decisions, as Chapters 6 and 10 explain.

Stability, heritability and change

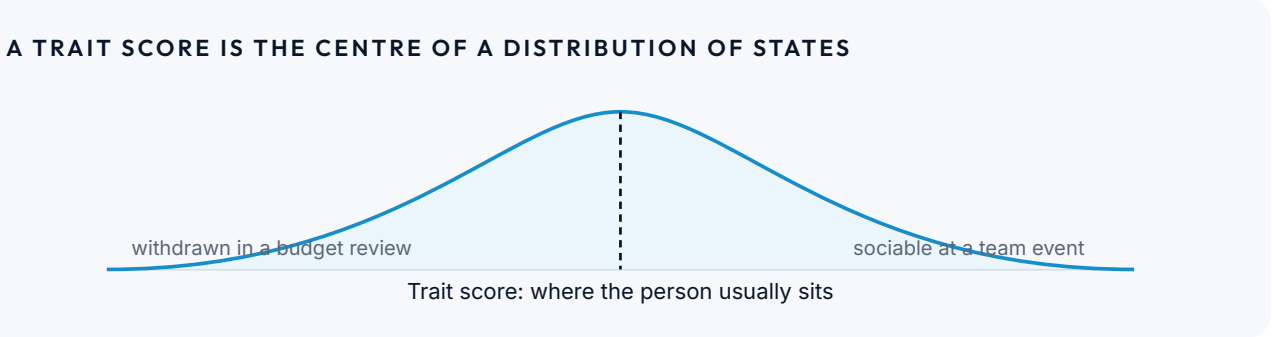
Big Five traits are moderately heritable, with twin studies typically estimating heritability around 40 to 50 per cent (Jang, Livesley and Vernon, 1996). Rank-order stability, meaning the tendency for people to keep their position relative to others, rises across adulthood and plateaus after about age 50 (Roberts and DelVecchio, 2000). Mean levels also change in predictable ways: on average, people become more conscientious, more agreeable and more emotionally stable as they age, a pattern called the maturity principle (Roberts, Walton and Viechtbauer, 2006).

TWO POINTS MATTER FOR ORGANIZATIONS

First, traits are stable enough to support decisions such as role fit. A profile measured this year will still describe the person reasonably well next year. Second, traits are not fixed. Work itself shapes personality over time, a finding examined in Chapter 4 (Holman and Hughes, 2021). This has a practical consequence: a profile should always carry a date, and organizations should expect profiles to shift, particularly after major changes in role, responsibility or workload.

Whole Trait Theory: traits as distributions

Fleeson's Whole Trait Theory reconciles stability and variability (Fleeson and Jayawickreme, 2015). A person does not behave at their trait level all the time. Instead, each person has a distribution of states, and their trait score describes the centre of that distribution. Someone with moderate Extraversion may be highly sociable at a team event and withdrawn in a budget review. Their trait score tells us where they usually sit, not where they sit at every moment.



This view matters for how reports are written. A trait score of 8 does not mean "always behaves at level 8". It means "tends to behave around this level, with variation depending on the situation". THE BRAIN's language rules (Chapter 7) therefore favour phrases such as "tends to" over "is". The distribution view is also central to the Contextual Switch interaction type (Chapter 5), which describes how certain situations shift a person's typical behaviour in predictable ways.

Why the Big Five suits organizations

Five properties make the Big Five the strongest available foundation.

Reliability	Established measures show good internal consistency and test–retest reliability.
Replication	The structure has been reproduced across many languages and cultures, though with variation (Chapter 3).
Prediction	Meta-analyses link the traits to job performance, leadership, teamwork and wellbeing (Chapter 4).
Hierarchy	Domains summarize; facets differentiate. Organizations can work at either level, depending on the decision.
Neutrality	Each pole of each trait has strengths. This supports development conversations rather than labelling.

The fifth property is easily underrated. Assessment that labels people as good or bad provokes defensiveness, invites gaming and damages trust. Assessment that describes trade-offs invites reflection. A low Extraversion score is not a deficit; it describes someone who may listen better, think more carefully before speaking and sustain focus on solitary work. Framing every trait as a trade-off is both more accurate and more useful.

CHAPTER 2 REFERENCES

Allport, G. W., & Odbert, H. S. (1936). Trait-names: A psycho-lexical study. *Psychological Monographs*, 47(1).

Costa, P. T., & McCrae, R. R. (1992). *Revised NEO Personality Inventory (NEO-PI-R) professional manual*. Psychological Assessment Resources.

Fleeson, W., & Jayawickreme, E. (2015). Whole Trait Theory. *Journal of Research in Personality*, 56, 82–92.

Goldberg, L. R. (1990). An alternative "description of personality": The Big-Five factor structure. *Journal of Personality and Social Psychology*, 59(6), 1216–1229.

Holman, D. J., & Hughes, D. J. (2021). Transactions between Big-5 personality traits and job characteristics across 20 years. *Journal of Occupational and Organizational Psychology*, 94(3), 762–788.

Jang, K. L., Livesley, W. J., & Vernon, P. A. (1996). Heritability of the Big Five personality dimensions and their facets. *Journal of Personality*, 64(3), 577–591.

Roberts, B. W., & DelVecchio, W. F. (2000). The rank-order consistency of personality traits from childhood to old age. *Psychological Bulletin*, 126(1), 3–25.

Roberts, B. W., Walton, K. E., & Viechtbauer, W. (2006). Patterns of mean-level change in personality traits across the life course. *Psychological Bulletin*, 132(1), 1–25.

CHAPTER 3

Competing Models and the Limits of Single Traits

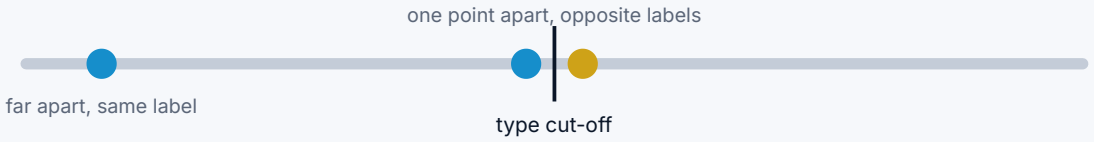
SUMMARY This chapter compares the Big Five with the models most often used in organizations, assesses each against scientific and practical criteria, and then turns to a limit that all trait models share: single traits explain only a modest share of behaviour. It reviews interactionist theory, from the person–situation debate to personality network models, as the foundation for THE BRAIN.

The models organizations actually use

Many organizations use type-based tools such as the Myers-Briggs Type Indicator (MBTI), DISC and the Enneagram. They are popular because they are easy to explain and give people a memorable label. A four-letter type or a single colour is easy to remember, easy to discuss in a workshop and easy to put on a name badge. These are real advantages, and any serious alternative has to take them seriously.

Their weaknesses are, however, well documented. Type models split continuous traits into categories, so two people a single point apart on a scale can receive opposite labels, while two people far apart on the same side receive the same label. MBTI shows weak test–retest consistency of type, with many people receiving a different type on retest (Pittenger, 2005). DISC and the Enneagram have limited peer-reviewed evidence of predictive validity for work outcomes. When a tool's labels change between sittings and do not predict outcomes, it may still be useful for starting conversations, but it should not inform decisions about people.

THE PROBLEM WITH CUT-OFFS



Research-grade alternatives are stronger. HEXACO adds a sixth factor, Honesty-Humility, which predicts counterproductive work behaviour and ethical conduct beyond the Big Five (Ashton and Lee, 2007). Cattell's 16PF offers finer detail at the cost of a less stable higher-order structure. The "Big Two" metatraits, Stability and Plasticity, sit above the five domains (Digman, 1997; DeYoung, 2006). Stability combines Conscientiousness, Agreeableness and Emotional Stability; Plasticity combines Extraversion and Openness. These higher-order factors are useful for broad summaries but lose the detail organizations need for specific decisions.

A comparative evaluation

MODEL	STRUCTURE	RELIABILITY EVIDENCE	WORKPLACE PREDICTION	CROSS-CULTURAL EVIDENCE	EASE OF USE
Big Five	5 continuous domains, 30 facets	Strong	Strong (meta-analytic)	Strong, with variation	High
HEXACO	6 continuous domains	Strong	Strong, especially for integrity	Good	Moderate
16PF	16 primary factors	Moderate to strong	Moderate	Moderate	Moderate
MBTI	16 types	Weak for type stability	Weak	Mixed	Very high
DISC	4 styles	Limited peer-reviewed evidence	Limited	Limited	Very high
Enneagram	9 types	Limited	Limited	Limited	High

Table 3.1 Qualitative comparison of personality models used in organizations. Ratings summarize the cited literature and are the authors' judgment.

The Big Five offers the best balance of evidence and usability. HEXACO's Honesty-Humility factor is a real addition, and organizations concerned with ethical conduct should take it seriously. THE BRAIN addresses it by including Integrity, Values and Ethics as separate traits outside the Big Five core (Chapter 6). This gives the framework the benefit of HEXACO's insight without abandoning the larger evidence base of the five-factor structure.

The usability of type models also carries a lesson. People want memorable patterns, not thirty numbers. THE BRAIN responds to this with archetypes (Chapter 8), but builds them on continuous scores and always reports them alongside the underlying profile. The label is an entry point, never a replacement.

The ceiling on single-trait prediction

Even the best trait model has a limit. Individual traits typically correlate with work outcomes at around r = .10 to .30, explaining a modest share of variance (Barrick, Mount and Judge, 2001; Roberts and others, 2007). Conscientiousness, the strongest single predictor of job performance, correlates at roughly r = .20 to .25 after corrections.

~6%
variation explained by r = .25

For readers less familiar with correlations: a correlation (r) runs from 0, meaning no relationship, to 1, meaning a perfect relationship. Squaring it gives the share of variation explained. A correlation of .25 explains about 6 per cent of the variation in performance. That sounds small, and for a single person it is a weak guide. Across hundreds of hiring decisions, however, it produces real and measurable gains, which is why personality assessment remains valuable despite modest correlations.

These are meaningful effects at organizational scale. But they leave most of the variance in behaviour unexplained. Part of the remainder is situational. Part is measurement error. And part, this monograph argues, may come from how traits combine.

A sobering test of the third possibility comes from Vize, Sharpe, Miller, Lynam and Soto (2022). In a registered report using samples of at least 1,350 per outcome, they tested all ten pairwise interactions of the Big Five domains against 81 life outcomes, 750 tests in total. Only 40 interactions (5.33 per cent) replicated robustly, all were small (around 1 per cent of additional variance), and adding interactions did not improve out-of-sample prediction.

750	81	5.33%	~1%
tests of pairwise interactions	life outcomes	replicated robustly (40)	additional variance each

Any framework built on trait interaction must answer this finding. Chapter 5 sets out why THE BRAIN's claims differ from what Vize and colleagues tested, and how they should be tested. Part V, Section C, examines the study in detail.

Interactionism: a century-old idea

The idea that personality is more than a list of separate traits is not new. Allport (1937) described personality as a dynamic organization of systems, not a list of parts. Murray (1938) stressed the interplay of needs and situational press, the demands and opportunities an environment presents. Mischel's (1968) critique of trait consistency triggered the person–situation debate, which ended in broad agreement: traits are real, but their expression depends on situations and on other characteristics of the person (Magnusson and Endler, 1977).

Mischel and Shoda (1995) formalized this in the Cognitive-Affective Processing System (CAPS). In CAPS, personality is a network of units such as goals, beliefs, expectations and emotional responses that activate and inhibit one another. Each person shows a stable signature of if–then patterns: if criticized publicly, then withdraw; if given autonomy, then excel. Two people may show the same average level of a behaviour but very different if–then signatures, and it is the signature that predicts what they will do in a particular situation.

CAPS is important for THE BRAIN in two ways. It shows that interactions within the person, between motives, beliefs and emotional tendencies, are a long-established part of personality theory. And it shows that those interactions are often triggered by situations. Several of the eleven interaction types, especially Contextual Switch and Masking, are close descendants of the CAPS idea.

Network models of personality

More recently, network psychometrics treats traits or items as nodes connected by edges that represent their conditional associations (Borsboom and Cramer, 2013; Costantini and others, 2015). In a network model, the question is not only how high someone scores on each trait, but how the traits are connected: which ones tend to rise and fall together once all the others are taken into account. Network studies show that the links between traits carry information beyond the trait levels themselves.

Network models are, however, mainly descriptive tools for researchers. They estimate associations across populations. They do not tell a manager what a specific combination means for a specific person. That gap between interactionist theory and applied interpretation is the problem THE BRAIN sets out to solve. It borrows the network idea, that relationships between traits matter, and turns it into a vocabulary practitioners can apply to individual profiles.

Lessons for an applied framework

Taken together, the competing models and the interactionist tradition suggest five design principles for any applied framework.

Continuous dimensions	Build on continuous, well-validated dimensions rather than types.
Specific traits	Add specific traits, such as integrity, where the evidence shows they predict beyond the core.
Memorable patterns	Offer memorable patterns for usability, but never let the pattern replace the profile.
Interactions and situations	Treat interactions and situations as part of personality, not as noise around it.
Honesty about evidence	Be honest that the predictive value of interactions is still an open empirical question.

THE BRAIN is designed around all five.

CHAPTER 3 REFERENCES

Allport, G. W. (1937). *Personality: A psychological interpretation*. Holt.

Ashton, M. C., & Lee, K. (2007). Empirical, theoretical, and practical advantages of the HEXACO model of personality structure. *Personality and Social Psychology Review*, 11(2), 150–166.

Barrick, M. R., Mount, M. K., & Judge, T. A. (2001). Personality and performance at the beginning of the new millennium. *International Journal of Selection and Assessment*, 9(1–2), 9–30.

Borsboom, D., & Cramer, A. O. J. (2013). Network analysis: An integrative approach to the structure of psychopathology. *Annual Review of Clinical Psychology*, 9, 91–121.

Costantini, G., and others (2015). State of the aRt personality research: A tutorial on network analysis of personality data in R. *Journal of Research in Personality*, 54, 13–29.

DeYoung, C. G. (2006). Higher-order factors of the Big Five in a multi-informant sample. *Journal of Personality and Social Psychology*, 91(6), 1138–1151.

Digman, J. M. (1997). Higher-order factors of the Big Five. *Journal of Personality and Social Psychology*, 73(6), 1246–1256.

Magnusson, D., & Endler, N. S. (Eds.). (1977). *Personality at the crossroads*. Erlbaum.

Mischel, W. (1968). *Personality and assessment*. Wiley.

Mischel, W., & Shoda, Y. (1995). A cognitive-affective system theory of personality. *Psychological Review*, 102(2), 246–268.

Murray, H. A. (1938). *Explorations in personality*. Oxford University Press.

Pittenger, D. J. (2005). Cautionary comments regarding the Myers-Briggs Type Indicator. *Consulting Psychology Journal*, 57(3), 210–221.

Roberts, B. W., Kuncel, N. R., Shiner, R., Caspi, A., & Goldberg, L. R. (2007). The power of personality. *Perspectives on Psychological Science*, 2(4), 313–345.

Vize, C. E., Sharpe, B. M., Miller, J. D., Lynam, D. R., & Soto, C. J. (2022). Do the Big Five personality traits interact to predict life outcomes? *European Journal of Personality*.

PART II

The BRAIN Framework

Chapter 4	What the Big Five Predicts at Work
Chapter 5	A Taxonomy of Trait Interactions
Chapter 6	The BRAIN Architecture: Trait DNA and the Interaction Matrix
Chapter 7	The Narrative Engine: A Constrained Reasoning Instrument

CHAPTER 4

What the Big Five Predicts at Work

SUMMARY This chapter reviews evidence linking the Big Five to job performance, leadership, teamwork and knowledge sharing, wellbeing and financial behaviour. It also covers the reciprocal relationship between traits and work, and the prediction of traits from digital behaviour. It ends with an honest account of effect sizes and what they imply for applied use.

Job performance

Barrick and Mount's (1991) meta-analysis of 117 studies (162 samples, N = 23,994) established Conscientiousness as the most consistent predictor of job performance across occupations. A meta-analysis combines the results of many studies into a single estimate, which is far more trustworthy than any one study alone. Later meta-analyses confirmed this finding and added nuance (Barrick, Mount and Judge, 2001). Emotional Stability predicts performance modestly across most jobs. Extraversion predicts performance in sales and management. Openness predicts training proficiency. Agreeableness predicts success in roles built on cooperation and service.

The pattern matters for design. No single trait is "good" everywhere. The value of a trait depends on role demands, which is why THE BRAIN gives each trait separate weights for hiring and for development (Chapter 6), and why Corporate Brain maps behavioural requirements to specific goals and functions rather than applying one ideal profile to everyone (Chapter 9).

TRAIT	WHERE IT PREDICTS MOST CLEARLY	PRACTICAL IMPLICATION
Conscientiousness	Almost all jobs and criteria	Give it high weight in most role profiles
Emotional Stability	Most jobs, modestly; high-pressure roles	Consider stress demands of the role
Extraversion	Sales, management, training	Weight it for client-facing and leadership roles only
Openness	Training and learning outcomes	Weight it for roles requiring rapid learning or innovation
Agreeableness	Service, teamwork, cooperation	Weight it for roles built on collaboration

Table 4.1 Where each domain predicts job outcomes, summarizing Barrick and Mount (1991) and Barrick, Mount and Judge (2001).

Leadership

Judge, Bono, Ilies and Gerhardt (2002) found that Extraversion was the strongest correlate of leadership emergence and effectiveness, followed by Conscientiousness, Openness and Emotional Stability. Agreeableness showed weaker links. The distinction between emergence and effectiveness matters. Emergence describes who gets seen as a leader; effectiveness describes who leads well. Extraversion is especially strong for emergence, which means organizations risk promoting confident, sociable people whether or not they lead well.

Leadership is a clear case where combinations matter. Extraversion without Emotional Stability or Integrity can produce a charismatic leader whose influence outruns their judgment. THE BRAIN treats this as a non-negotiable flag, Charismatic Risk, triggered by high Persuasive Influence alongside low Integrity. Conversely, Leadership with Empathy, Leadership with Accountability and Leadership with Emotional Stability are among the most valuable amplification patterns in the framework.

Leadership capability is not one trait. It is a configuration.

Teamwork and knowledge sharing

Agreeableness and Extraversion support cooperation, helping and information exchange. Studies of knowledge sharing, including online settings, generally find Extraversion to be the most consistent predictor. One recent study of Thai social-commerce users reported Extraversion and Neuroticism as predictors of knowledge sharing (Phuthong, 2023), though its measurement limits mean its specific figures should be treated with caution (see Part V, Section D).

At team level, a team's average Conscientiousness and Agreeableness, and its lowest member's scores, predict team performance (Bell, 2007). The finding about the lowest member is worth dwelling on. In tasks where everyone's contribution is needed, one person's low conscientiousness can hold back the whole team. This is a team-level version of the Ceiling Effect described in Chapter 5, and it suggests that team composition should be analysed not only by averages but by the gaps.

Wellbeing and social support

Personality is among the strongest correlates of subjective wellbeing (DeNeve and Cooper, 1998). In a sample of 848 university students, Yu, Chang and Li (2024) found that personality explained a large share of wellbeing variance, with Conscientiousness most strongly associated. Social support carried part of the effect for Extraversion and Agreeableness. For organizations, the implication is that support structures matter most for the employees whose wellbeing depends most on relationships, while others benefit more from autonomy, structure and the ability to manage their own work well.

Wellbeing findings are used in THE BRAIN only for development and support. They play no part in hiring, as Chapter 10 explains.

Financial behaviour

Campbell, Exley and Doyle (2023) review evidence that Conscientiousness relates to higher income, net worth and financial literacy, that Neuroticism relates to lower wealth and greater risk aversion, and that Agreeableness relates to lower income. These findings inform the financial goal mapping in Chapter 9 and our related work on financial behavioural assessment. The review's authors disclose links to a commercial assessment provider, which readers should weigh, and key claims should be checked against their primary sources.

Traits and work shape each other

Traits influence the jobs people choose and how they perform. Work also changes traits. Holman and Hughes (2021) followed 1,049 American employees across three surveys spanning 20 years. Sustained high workload led to increases in Openness, Extraversion and Agreeableness; job discretion had no effect on personality; and personality had little effect on later job characteristics, except that Neuroticism predicted greater job discretion. This supports a developmental view: organizations do not simply select personalities, they also shape them.

The practical message is twofold. Organizations should treat profiles as current tendencies rather than permanent facts, and should reassess at sensible intervals. And organizations bear some responsibility for the personalities they help to form. The roles, demands and resources an organization provides will, over years, leave a mark on the people who fill them.

Predicting traits from digital behaviour

Kosinski, Stillwell and Graepel (2013) showed that Facebook Likes predicted Big Five traits and other private attributes with accuracy comparable to some questionnaire measures. Workplace equivalents now exist: patterns in email, chat, calendars and project tools carry personality signal. THE BRAIN's behaviour mapping (Chapter 8) builds on this idea. The same finding raises serious privacy concerns, because the methods that infer personality from behaviour can also infer characteristics that must remain private. Chapter 10 sets out the safeguards.

What the effect sizes mean

Most single-trait correlations with work outcomes fall between r = .10 and .30. Across hundreds of decisions, such effects add real value. For any one person, however, a single trait is a weak guide. Two responses follow. One is humility: assessment should inform, not decide. The other is to look for more information in the profile, especially in how traits combine. That is the subject of Chapter 5.

EXPLAINING EFFECT SIZES TO MANAGERS

Personality scores shift the odds, they do not settle the outcome. If an assessment raises the chance of a good hire from 50 per cent to 60 per cent, it is valuable across a hundred hires and still wrong about many individuals. That is why the final judgment always rests with a person who knows the role and has met the candidate.

CHAPTER 4 REFERENCES

Barrick, M. R., & Mount, M. K. (1991). *Personnel Psychology*, 44(1), 1–26.

Barrick, M. R., Mount, M. K., & Judge, T. A. (2001). *International Journal of Selection and Assessment*, 9(1–2), 9–30.

Bell, S. T. (2007). Deep-level composition variables as predictors of team performance: A meta-analysis. *Journal of Applied Psychology*, 92(3), 595–615.

Campbell, Exley, & Doyle (2023). The Big Five personality traits (OCEAN) and financial planning. *Financial Services Review*.

DeNeve, K. M., & Cooper, H. (1998). The happy personality. *Psychological Bulletin*, 124(2), 197–229.

Judge, T. A., Bono, J. E., Ilies, R., & Gerhardt, M. W. (2002). Personality and leadership. *Journal of Applied Psychology*, 87(4), 765–780.

Kosinski, M., Stillwell, D., & Graepel, T. (2013). Private traits and attributes are predictable from digital records of human behavior. *PNAS*, 110(15), 5802–5805.

Phuthong, T. (2023). OCEAN traits as key determinants of knowledge sharing. *Journal of Human, Earth, and Future*.

Yu, Chang, & Li (2024). *Education Sciences*, 14, 1050.

CHAPTER 5

A Taxonomy of Trait Interactions

SUMMARY This chapter presents the theoretical core of the monograph: a taxonomy of eleven ways in which two traits can influence each other's behavioural expression. Each type is defined by its score condition, its mechanism, its observable signature and the response it calls for in assessment. The taxonomy is offered as a contribution to personality psychology that stands independently of its use in THE BRAIN.

Why interaction is the unit of behaviour

Statistical research usually treats traits as separate predictors whose effects add up. Interactionist theory (Chapter 3) suggests behaviour often works differently. One trait can strengthen, weaken, hide, block or redirect another. Applied practice has had only two words for trait relationships: correlated or independent. Neither says what a combination does.

The taxonomy below supplies that missing vocabulary. It was developed through expert analysis of trait pairs in organizational assessment, informed by the interactionist literature. Its status is theoretical: each type is a testable proposal, and the final section of this chapter sets out how to test them.

The evidential status of the taxonomy must be stated plainly. When broad Big Five domains are tested as simple statistical products against general life outcomes, interactions are rare and small (Vize and others, 2022). THE BRAIN differs from that test in four ways: it works with 91 narrower traits rather than five domains; it specifies 58 priority pairs chosen on theoretical grounds rather than all possible pairs; it defines interactions by score thresholds rather than linear products; and it targets specific behaviours in specific situations, such as decisions under pressure, rather than general life outcomes. Each difference is a reason the finding may not transfer. None is yet evidence that it does not. The eleven types are therefore offered as an interpretive taxonomy whose predictive value is a hypothesis for testing.

Why, then, use the taxonomy before that evidence is in? Because practitioners already reason about combinations, informally and inconsistently. A manager who says "she's ambitious, but she loses it under pressure" is describing a Conflict pattern. A coach who says "he'd be a great communicator if he ever listened" is describing a Ceiling Effect. The taxonomy does not invent this reasoning. It names it, makes it consistent, and makes it testable.

Four elements of every type

Score condition
The score pattern under which it applies, on the STEN scale (high = 7 to 10, low = 1 to 4).

Mechanism
How one trait changes the other's expression.

Behavioural signature
What observers actually see.

Assessment response
What a practitioner should do with it.

THE TAXONOMY

The eleven types at a glance

TYPE	SCORE CONDITION	MECHANISM	EXAMPLE PAIR
Amplification	Both high	Each enables the other; the combined effect exceeds either alone	Ambition × Result Orientation
Compensation	One high, one low	The high trait creates a workaround for the low trait's risk	Self-Regulation (high) × Ambition (low)
Conflict	One high, one opposing trait low	Competing drives produce visible inconsistency	Ambition (high) × Self-Regulation (low)
Suppression	One high, other mid or high	The dominant trait silently prevents the other's expression	Accommodating × Accountability
Paradox	Two opposing traits both high	A genuine duality, deployed situationally	Authoritarian × Democratic Leadership
Dependency	One high, enabler below threshold	The high trait needs an enabler to work as intended	Initiative × Growth Mindset
Ceiling Effect	One high, a limiting trait low	The low trait caps how far the high trait can go	Communication (high) × Active Listening (low)
Masking	One high, a concerning score elsewhere	A real strength hides a risk from observers and the person	Mental Toughness × wellbeing distress indicators
Contextual Switch	Both moderate to high	The relationship reverses under a trigger such as pressure	Empathy × Dominance
Compounding Development	Three or more growth traits high	Growth traits feed each other into self-sustaining development	Growth Mindset × Initiative × Hardiness
Risk Multiplier	One high, a risk trait low	The strength becomes the channel through which the risk spreads	Persuasive Influence × low Integrity

Table 5.1 The eleven interaction types.

The eleven types in detail

For each type we describe the mechanism, what it looks like at work, how a practitioner can recognize it in conversation, and what to do about it.

Amplification Both high

When two traits are both high, each can create conditions for the other to express more fully. Ambition supplies drive; Result Orientation supplies the delivery structures that turn drive into outcomes. Together they produce someone who builds systems to guarantee results, which neither trait produces alone. Amplification is the most common type among expert-defined pairs (33 of 58). The most valuable amplifications produce behaviour that is both rare and organizationally important, such as Leadership with Empathy, or Integrity with Accountability.

What it looks like. Amplified strengths tend to be the things colleagues mention first when describing someone. They are consistent, visible and reliable across situations. The person often does not experience them as effortful.

Recognizing it. Ask for examples of the person's best work. Amplified pairs usually appear together in those stories: the ambitious goal and the system built to hit it; the honest conversation and the ownership of what followed.

ILLUSTRATIVE COMPOSITE

A regional operations lead scores 9 on Accountability and 8 on Integrity. Her team describes her as the person who "says what she'll do and then does it, including the bad news". When a supplier error caused a missed deadline, she told the client before they noticed, explained the fix and absorbed the criticism herself. The pattern, Principled Accountable, is the foundation of trust in her region.

Assessment response. Name the combined strength and design roles that use it. Amplified pairs are the natural basis for role design, stretch assignments and succession planning. Watch also for the shadow side: some amplifications, such as Result Orientation with Ambition, can overrun quality or relationships when pushed hard.

Compensation One high, one low

A high trait can offset the risk of a related low trait without removing it. High Self-Regulation with low Ambition does not produce a disengaged employee. It produces a disciplined, reliable performer who delivers well within a defined scope but will not set stretch goals unprompted. Compensation shows why reading a low score in isolation can mislead: a practitioner who sees low Ambition without checking for compensating partners may underrate a dependable contributor.

What it looks like. Compensated weaknesses are often invisible in everyday work. The risk appears only when the compensating trait is overloaded or absent, for example when a disciplined but unambitious employee is given an open-ended goal with no structure.

Recognizing it. When a low score seems inconsistent with the person's track record, look for a compensating trait. Ask how the person handles situations that would normally expose the weakness.

Assessment response. Check every low score for compensating partners before treating it as a gap. In development, protect the compensating trait: it is doing important work.

Conflict

One high, one opposing trait low

Conflict arises when a high trait and an opposing low trait compete, producing behaviour others experience as inconsistent. High Ambition with low Self-Regulation is the paradigm case: brilliant in calm periods, reactive and damaging under pressure. Our practice suggests this is a common pattern among derailing high performers. It is why the Drive × Regulation domain crossing carries the highest multiplier in THE BRAIN (Chapter 6).

What it looks like. Colleagues describe the person in contradictory terms: "inspiring but unpredictable", "a star when things go well". Performance reviews show high peaks and sharp troughs.

Recognizing it. Ask about the last time things went badly under time pressure. Listen for reactive decisions, regret, and damage to relationships that had to be repaired afterwards.

ILLUSTRATIVE COMPOSITE

A business development manager with Ambition 9 and Self-Regulation 3 consistently exceeds quarterly targets. In the final week of each quarter, however, he discounts aggressively, promises delivery dates operations cannot meet and argues publicly with colleagues who push back. His results are excellent; the cost of his results is borne by everyone around him. THE BRAIN names this pattern Reactive Achiever and raises it as a high-priority flag.

Assessment response. Flag as a priority development need; probe pressure behaviour in interviews. Development focuses on the low trait, using concrete rehearsal of trigger situations rather than general advice to "stay calm".

RESEARCH IN PROGRESS

We are collecting data on how often this pattern occurs in assessed populations and how it relates to independent derailment indicators. Until that evidence is available, the claim that Conflict patterns predict derailment is based on practitioner experience, not measured outcomes.

Suppression

One high, other mid or high

Suppression differs from Conflict in being invisible. The suppressed trait is present but does not reach behaviour, because the dominant trait acts first. A highly accommodating person may genuinely value standards yet never hold others to account, because the drive for harmony always wins. The person experiences no tension; they may describe it as a natural preference. A mid-range Accountability score gives no warning unless the practitioner looks at what sits beside it.

What it looks like. Standards drift in the person's area without obvious conflict. The team likes them; performance problems persist.

Recognizing it. Ask what the person does when a well-liked colleague repeatedly misses a deadline. Suppression shows up as sincere commitment to standards in principle, combined with no examples of enforcing them in practice.

Assessment response. Look behind mid-range scores for a dominant suppressor. Development makes the suppressed behaviour easier: scripts for difficult conversations, agreed escalation rules, and explicit permission to prioritize standards over harmony.

Paradox Two opposing traits both high

When two conceptually opposing traits are both high, the result is not confusion but a genuine dual capacity. A leader high on both Authoritarian and Democratic styles is directive in crises and consultative in planning. The difficulty lies with observers who cannot see the contextual trigger and experience the leader as unpredictable.

What it looks like. Strong performance across very different situations, combined with team feedback that the leader is "hard to read".

Recognizing it. Ask the person how they decide when to consult and when to decide alone. Leaders with a true Paradox profile usually have a clear, if unspoken, rule.

Assessment response. Make the triggers explicit to the person and their team. A leader who says "in a crisis I will decide quickly and explain later; in planning I want everyone's view" turns apparent unpredictability into a trusted pattern.

Dependency One high, enabler below threshold

Some traits need an enabler. High Initiative without Growth Mindset produces restless, repetitive action: many starts, little learning. Growth Mindset turns each action into a development event. The practical insight is efficiency: developing the enabler unlocks a strength the person already has.

What it looks like. Lots of energy without a matching upward trajectory. The person launches initiatives, but the fifth looks much like the first.

Recognizing it. Ask what the person learned from their last failed initiative and what they did differently next time.

Assessment response. Invest in the enabler, not the dependent trait. Telling a high-Initiative person to take more initiative is wasted effort; helping them build a habit of structured reflection can transform their contribution.

READING THE TYPES

Each type is specified by its score condition, its mechanism, its behavioural signature and the assessment response it calls for. Table 5.1 summarizes all eleven.

Ceiling Effect One high, a limiting trait low

A low trait can cap a high one. An excellent communicator with very low Active Listening transmits brilliantly but cannot receive. At a certain level of relationship depth or leadership complexity, people stop trusting the communication because they feel unheard. Developing Communication further will not help. The return lies in raising the ceiling trait.

What it looks like. Early success that plateaus, often at the point where the role shifts from individual contribution to influence over others.

Recognizing it. Ask colleagues, not the person, whether they feel heard. Ceiling Effects are often invisible from the inside.

Assessment response. Locate the ceiling trait and target development there. This is often the highest-return development investment available for an otherwise strong performer.

Masking One high, a concerning score elsewhere

Masking requires the most care. A real, functional strength hides a concern from observers and sometimes from the person. Very high Mental Toughness alongside elevated distress indicators produces someone who keeps performing, does not show strain and does not seek support. The risk accumulates unseen until it surfaces suddenly.

What it looks like. Consistent performance, no visible complaints, and then an abrupt change: sudden resignation, illness, or a sharp drop in engagement that seems to come from nowhere.

Assessment response. Treat as a confidential wellbeing priority, never a performance judgement. It is surfaced only to the individual and, with consent, to a designated wellbeing contact. It is never used in hiring decisions (Chapter 10). A strong, stable presentation is not a reason to withhold support; it may be the reason support is needed.

Contextual Switch Both moderate to high

Some pairs reverse under specific triggers such as high stakes, public failure or time pressure. Empathy with Dominance at moderate-high levels produces a warm, decisive leader at baseline. Under pressure, empathy, which costs more effort, drops away, and dominance takes over. The team experiences two different leaders. The switch is involuntary. Whole Trait Theory (Chapter 2) explains it as a shift in the person's state distribution under certain conditions.

What it looks like. Team members describe the leader differently depending on when they joined or which projects they worked on. Engagement surveys diverge between calm and crisis periods.

Recognizing it. Ask the person and their team to describe the leader on a normal day and on the worst day of the last quarter.

Assessment response. Identify the trigger conditions and plan for them. Mental rehearsal of trigger situations, agreed signals with trusted colleagues, and small structural changes such as a pause before responding to escalations can all reduce the switch.

Compounding Development

Three or more growth traits high

When three or more growth-enabling traits are high (for example Growth Mindset, Initiative, Hardiness, Resilience, Learning Attitude and Adaptability), development becomes self-sustaining. Each trait feeds the others. Standard supportive coaching underuses this person; stretch assignments serve them better.

What it looks like. Rapid, visible growth across successive roles. The person seeks feedback, recovers from setbacks quickly and applies lessons to new problems.

Assessment response. Offer stretch challenges, not support programmes. The main risk is under-use: people with this pattern may leave if they stop growing.

Risk Multiplier

One high, a risk trait low

The inverse of Amplification. A real strength becomes the channel through which a risk operates faster and at greater scale. High Persuasive Influence without Integrity is the clearest case: the person can make self-serving decisions look principled and rally others around poor choices. This is more dangerous than low integrity alone because it is harder to detect.

What it looks like. Persuasive success combined with recurring patterns of decisions that benefit the person more than the organization, often defended convincingly.

Assessment response. Escalate; require an ethics-focused structured interview in hiring or senior development. Never resolve a Risk Multiplier flag on the strength of a confident interview alone, because confident interviews are exactly what this pattern produces.

How the types relate to each other

The eleven types form natural pairs and families. Amplification and Risk Multiplier are mirror images: one magnifies a strength, the other magnifies a risk. Conflict and Suppression both involve a dominant and a weaker trait, but one is visible and the other hidden. Dependency and Ceiling Effect both involve a limiting trait, but one is an enabler that is missing and the other a cap that is low. Compensation and Masking both involve a strength covering a weakness, but compensation manages the risk while masking conceals it.

A useful way to organize the types is along two axes: whether the interaction is visible to observers, and whether it enhances strength or risk.

	VISIBLE TO OBSERVERS	HIDDEN FROM OBSERVERS
STRENGTH-ENHANCING	Amplification, Paradox, Compounding Development	Compensation, Dependency (when the enabler is present)
RISK-ENHANCING	Conflict, Ceiling Effect, Contextual Switch	Suppression, Masking, Risk Multiplier

Table 5.2 The eleven types arranged by visibility and direction.

The hidden, risk-enhancing quadrant is where systematic assessment adds the most value.

Visible risks are usually noticed eventually by managers and colleagues. Hidden risks, by definition, are not, and they tend to surface late and expensively.

Relation to existing concepts

Several types correspond to established statistical ideas. Amplification and Risk Multiplier resemble positive and negative moderation, where one variable changes the strength of another's effect. Suppression parallels the statistical suppressor effect. Contextual Switch resembles three-way interactions involving situational moderators. The taxonomy's contribution is not to replace these concepts but to translate them into behavioural terms that practitioners can use, and to add types, such as Masking and Ceiling Effect, that have no simple statistical counterpart.

Testing the taxonomy

The taxonomy makes testable predictions. For each type, a study can compare a model with only the two traits' main effects against a model that adds their interaction term, predicting a relevant outcome such as manager-rated performance, derailment indicators or engagement.

RESEARCH IN PROGRESS

The first tests will focus on five Tier 1 pairs, one per type family:

CONFLICT	SUPPRESSION	CEILING EFFECT	DEPENDENCY	RISK MULTIPLIER
Ambition × Self-Regulation	Accommodating × Accountability	Communication × Active Listening	Initiative × Growth Mindset	Persuasive Influence × Integrity

For each, we will report sample size, outcome measure, variance explained by main effects, additional variance from the interaction, and significance.

Tests will follow the standards set by Vize and colleagues (2022): pre-registration of hypotheses; a stated smallest effect of interest (about 1 per cent of additional variance); samples large enough to detect it, roughly 1,000 to 1,500 participants; split-sample replication; outcomes rated by others rather than self-reported where possible; and an out-of-sample check that a model including Tier 1 interactions predicts held-out outcomes better than main effects alone. These samples will require pooling consented data across client organizations.

A positive result for even a few types would greatly strengthen the framework. A negative result would also be valuable: it would tell us which types should remain interpretive tools only, and which should be revised or retired.

CHAPTER 5 REFERENCES

Aiken, L. S., & West, S. G. (1991). *Multiple regression: Testing and interpreting interactions*. Sage.

Fleeson, W., & Jayawickreme, E. (2015). *Journal of Research in Personality*, 56, 82–92.

Vize, C. E., Sharpe, B. M., Miller, J. D., Lynam, D. R., & Soto, C. J. (2022). *European Journal of Personality*.

Sharma, S., & Kumar, S. (in preparation). THE BRAIN: Behavioural Reasoning and Interaction Network.

CHAPTER 6

The BRAIN Architecture: Trait DNA and the Interaction Matrix

SUMMARY This chapter describes the first two layers of THE BRAIN. The Trait DNA layer profiles 91 traits across ten dimensions. The Interaction Matrix classifies all 4,095 trait pairs into three tiers of significance. The chapter also sets out the non-negotiable flags, the Consolidation Model that combines technical and behavioural information, and the safeguards that keep wellbeing data out of hiring decisions.

Overview

THE BRAIN has three layers that run in sequence. Layer 1, Trait DNA, describes what each trait means in behaviour. Layer 2, the Interaction Matrix, describes what each pair of traits does together. Layer 3, the Narrative Engine (Chapter 7), applies both to an individual's scores and writes the report. Every statement in a final report can be traced back through this chain to a score, an interaction type and a rule.

LAYER 1
Trait DNA
What each trait means in behaviour

LAYER 2
Interaction Matrix
What each pair of traits does together

LAYER 3
Narrative Engine
Applies both to a profile and writes the report

STEP	LAYER	INPUT	OUTPUT
1	**Flag check**	STEN scores	Any non-negotiable flags, placed first
2	**Trait DNA**	STEN scores	Behavioural meaning of each high and low score
3	**Interaction Matrix**	Dominant and suppressed traits	Live interactions by tier and type
4	**Narrative Engine**	Ranked interactions and archetype	Written report
5	**Quality gate**	Draft report	Released report, or return to step 4

Table 6.1 The flow from scores to report.

LAYER 1

Trait DNA

The inventory spans five main groups: Personality, Culture, Skill, Ability and Interest, plus smaller Wellness and Parenting groups used in specific products.

GROUP	EXAMPLE TRAITS	DOMAINS
Personality	Big Five, Motivation and Drive, Interpersonal, Intrapersonal, Leadership Styles, Thinking Styles	`BIG5, DRIVE, INTER, INTRA, LEAD, THINK`
Skill	Leadership and Influence, Communication, Execution, Client and Customer, Professional	`COMM, EXEC, CX, PROF`
Ability	Cognitive Abilities, Learning Styles, Study Skills	`COG, LSTYLE, LSTYLE2, STUDY`
Culture	Values and Ethics, Organizational Culture	`VALUES, ORG`
Interest	Domain interests	`INT`
Wellness	Wellbeing indicators	`WELL`
Parenting	Context-specific traits	`PARENT`

Table 6.2 Trait inventory by group. The full list of 91 traits by domain appears in Appendix A.

Each trait is profiled on ten dimensions, including:

High- and low-score behaviour	STEN 7 to 10 and STEN 1 to 4, written as consequences rather than definitions. Integrity, for example, is described at the high end as consistent alignment between values and actions and honesty under pressure, not simply "honest".
Top interaction partners	The traits it most meaningfully combines with.
Acquisition weight (1 to 5)	Its priority in hiring.
Development weight (1 to 5)	Its priority in coaching.
Risk flag condition	The score pattern that triggers a flag.
Domain code	Used to apply domain multipliers.
Scoring direction	Showing whether a high score is the adaptive or the risk pole.

The dual weighting reflects a key idea: the same trait matters differently depending on the decision. Growth Mindset carries an acquisition weight of 4 but a development weight of 5, because it is the best single signal of whether development investment will pay off. Openness carries a development weight above its acquisition weight, reflecting its link to training proficiency. Conscientiousness is weighted highly for both, reflecting its general predictive value.

Why write behaviour, not definitions

A definition tells a reader what a trait is. A behavioural description tells them what to expect. Compare "Self-Regulation: the capacity to control impulses" with "At STEN 3, Self-Regulation creates a specific vulnerability under pressure: reactive decisions that damage relationships when it matters most." The second is longer, but a manager can act on it. Writing every trait description this way forces precision about what the trait does at work, and makes it easier to test whether descriptions match observed behaviour.

LAYER 2

The Interaction Matrix

With 91 traits there are 4,095 possible pairs. Specifying every pair by hand is neither feasible nor useful. The matrix therefore concentrates expert effort where consequences are highest.



TIER	PAIRS	SHARE	WHO DEFINES IT	HOW IT IS USED
Tier 1: expert-defined	58	1.4%	Human experts	Used directly in report narratives
Tier 2: rule-generated	2,394	58.5%	Domain rule engine	Supporting context, moderate-depth insight
Tier 3: engine-inferred	1,643	40.1%	Engine reasoning from Trait DNA	Inferred only when relevant to a profile
Total	4,095	100%		

Table 6.3 Interaction Matrix tiers.

Tier 1

Tier 1 pairs meet at least one of three criteria: one trait lists the other as a top partner; the pair crosses domains with a multiplier of 1.3 or higher; or a risk flag involves both traits. Each Tier 1 pair has a full specification: interaction type, pattern name, behavioural insight, risk or watch note, hiring note, development note and multiplier. Their type distribution is shown below, and the complete catalogue appears in Appendix F.

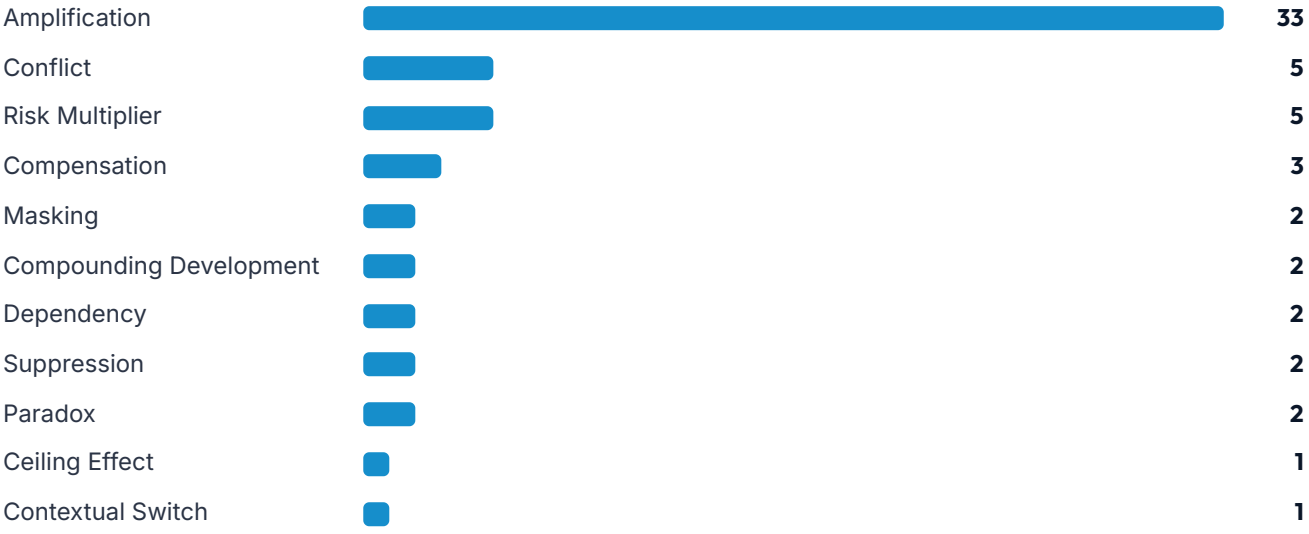


Table 6.4 Tier 1 pairs by interaction type.

Tier 2

Tier 2 pairs are generated by domain rules. Each domain crossing has an expected interaction pattern and a multiplier. The highest is Drive × Regulation at 1.5×, reflecting the central importance of how motivational intensity meets self-control. The full rule weights are proprietary; the principle and the most common patterns are published here.

TIER 2 PATTERN	PAIRS	TYPICAL MEANING
Cross-Domain Contribution	1,975	Traits from different domains that together broaden contribution
Domain Reinforcement	139	Two traits in the same domain strengthening a shared tendency
Values-anchored Performance	48	Values traits channelling performance traits
Regulated Relationship	47	Self-regulation shaping interpersonal behaviour
Personality-Relational Fit	35	Core personality supporting relationship traits
Cognitive Executor	35	Cognitive ability converted into execution
Drive-Regulation Tension	34	Motivational drive meeting self-control
Leader-Executor	30	Leadership traits combined with delivery traits
Relational Communicator	27	Interpersonal traits combined with communication
Thinking Executor	24	Thinking style combined with execution

Table 6.5 The ten most common Tier 2 patterns (from the interaction matrix).

Part V, Section H, discusses a known limitation: most Tier 2 pairs currently share one generic pattern, and the rule engine is being revised to make Tier 2 rules conditional on score patterns and scoring direction.

Tier 3

Tier 3 pairs mostly involve Interest, Wellness, Parenting and Learning Style traits in low-weight cross-domain combinations. When one becomes relevant to a profile, the narrative engine reasons about it from the two traits' DNA profiles. Because these inferences are produced at the time a report is written, they are logged, and the most frequently inferred pairs are reviewed by experts for promotion to Tier 1 or Tier 2.

The design principle is simple: put the intelligence where behavioural consequence is highest. Fewer than 1.5 per cent of pairs are hand-specified, yet we expect them to drive most of the insight in a typical report.

RESEARCH IN PROGRESS

We are measuring the share of report statements drawn from Tier 1, Tier 2 and Tier 3 across a sample of generated reports, to test whether expert-defined pairs do carry most of the interpretive weight.

SAFEGUARDS

Non-negotiable flags

Twelve flag conditions are checked before any other analysis. They ensure that critical concerns are never buried in a narrative.

FLAG	PRIORITY	USE IN HIRING	USE IN DEVELOPMENT
Wellbeing Concern	Immediate	Not used. Triggers a confidential referral only	Pause other development; offer wellbeing support
Integrity Deficit	Critical	Flag to hiring manager; structured ethics interview	Priority values coaching
Accountability Deficit	Critical	Flag for interview probing	Priority coaching
Reactive Achiever	High	Probe pressure behaviour	Self-regulation development
Power-Blind Authority	High	Probe	Leadership coaching
Charismatic Risk	High	Probe	Leadership coaching
Masking Risk	High	Not used. Confidential referral only	Wellbeing support
Values Deficit (general)	High	Probe	Values coaching
Reckless Innovator	Moderate	Note	Risk-judgment development
Motivated but Fragile	Moderate	Note	Resilience support
Delegation Risk	Moderate	Note	Delegation coaching
Conflict-Avoidant Loyalist	Moderate	Note	Assertiveness coaching

Table 6.6 Non-negotiable flags and how they are used. Flags never produce automatic rejection; they prompt human review.

WELLBEING SAFEGUARD

Two flags, Wellbeing Concern and Masking Risk, draw on indicators of distress, including depression-related items. In THE BRAIN as described here, these indicators are excluded entirely from hiring and screening. In a hiring context they are not scored for decision purposes and are not shown to the hiring manager. In a development context, results go first to the individual, and onward to a designated wellbeing contact only with the individual's consent. This rule responds to disability-discrimination law and to the ethical principle that health information should support people, not filter them out (Chapter 10).

The Consolidation Model

Organizations often need one view that combines technical competence with behavioural profile. THE BRAIN's Consolidation Model does this:

$$\text{Consolidated score} = (T \times w_T) + (B \times w_B)$$

where $w_T + w_B = 100\%$. T is the technical score, B the purpose-weighted behavioural score, and the weights are set by the client for each role.

A software engineering role might use 70 per cent technical and 30 per cent behavioural; a people-leadership role might reverse this. The weights are a deliberate, visible choice made by the organization, not hidden inside the system.

In development contexts, a wellbeing sustainability indicator is reported alongside the consolidated score. It is built from eight indicators, including Emotional Stability, Stress Tolerance, Resilience and Self-Care. It is not multiplied into the score and is never applied in hiring.

The consolidated score, read with its two components, places a person in one of four quadrants.

	BEHAVIOURAL HIGH	BEHAVIOURAL LOWER
TECHNICAL HIGH	**Deploy** Ready for responsibility in roles that use both strengths	**Manage** Strong contributor who needs behavioural support or careful role design
TECHNICAL LOWER	**Develop** Strong behavioural base; invest in technical capability	**Review** Discuss fit and support carefully, with human judgment

Table 6.7 Consolidation quadrants. Cut-offs are set per role by the client.

CHAPTER 6 REFERENCES

Bakker, A. B., & Demerouti, E. (2017). Job demands–resources theory: Taking stock and looking forward. *Journal of Occupational Health Psychology*, 22(3), 273–285.

Hakanen, J. J., & Schaufeli, W. B. (2012). Do burnout and work engagement predict depressive symptoms and life satisfaction? *Journal of Affective Disorders*, 141(2–3), 415–424.

CHAPTER 7

The Narrative Engine: A Constrained Reasoning Instrument

SUMMARY This chapter describes Layer 3 of THE BRAIN, the Narrative Engine. It distinguishes a constrained reasoning instrument from a free-form writer, sets out the engine's eight-step protocol and language rules, and explains how traceability and quality checks keep reports accurate and accountable. It closes with what the system does and does not claim.

Two ways to automate interpretation

There are two broad ways to automate the writing of assessment reports. The first is free-form generation: give a general-purpose writing system a profile and ask it to describe the person. This produces fluent text quickly, but it carries well-known problems. Such systems can state false things confidently, give different answers to the same input, and draw on general patterns that have nothing to do with the assessment science behind the scores.

THE BRAIN takes the second route. It does not ask the engine to produce psychological insight from general knowledge. It gives the engine a fixed knowledge base (the Trait DNA, the Interaction Matrix and the flag rules) and a fixed reasoning procedure, and asks it to apply them to one profile at a time. **The engine is the instrument; the intelligence sits in the framework.** Experts write the rules; the engine applies them consistently.

The eight-step protocol

1	**Flag check**	Test all non-negotiable conditions first. A triggered flag takes precedence over everything else in the report. In hiring mode, wellbeing indicators are excluded before this step.
2	**Score sorting**	Separate dominant traits (STEN 7 or above), suppressed traits (STEN 4 or below) and mid-range traits.
3	**Live interaction search**	For each dominant trait, look up its partners in the matrix. A partner in the suppressed list suggests Conflict, Suppression, Ceiling Effect or Risk Multiplier. A partner in the dominant list suggests Amplification or Paradox. A partner in the mid-range suggests Dependency.
4	**Ranking**	Weight each live interaction by tier, domain multiplier and the purpose weight (acquisition or development) of both traits. Keep the most significant.
5	**Archetype match**	Identify the Big Five archetype and any modifiers (Chapter 8).
6	**Narrative generation**	Write the report around the ranked interactions.
7	**Language rules**	Apply the language rules set out below.
8	**Quality check**	Test the draft against five criteria before release (see Quality checks). A

A worked example

ILLUSTRATIVE COMPOSITE

A candidate for a regional sales manager role has these selected scores:

Ambition 9 **Result Orientation 8** **Persuasive Influence 8** **Integrity 8** **Empathy 4**
Active Listening 3 **Self-Regulation 6**

Step 1 finds no flags: Integrity is high, so Charismatic Risk does not trigger, and Self-Regulation is mid-range, so Reactive Achiever does not trigger. Step 2 sorts Ambition, Result Orientation, Persuasive Influence and Integrity as dominant; Empathy and Active Listening as suppressed. Step 3 finds several live pairs: Result Orientation × Ambition (Amplification, High-Drive Achiever); Result Orientation × Integrity (Amplification, Principled Executor); Result Orientation × Empathy (Conflict, Hard-Driving but Blind); and, because Communication is also mid-high, a possible Active Listening ceiling. Step 4 ranks the Conflict and the Ceiling Effect highly for a people-management role because both involve traits with high acquisition weights for leadership. The final report opens with the candidate's strong, principled drive, and then states clearly that relationships with the team and clients may suffer under target pressure, citing the scores behind each claim, and recommends interview questions on how the candidate handled a team member who was struggling.

Language rules

The engine's writing rules matter as much as its logic.

Consequence, not description
Not "low score on Self-Regulation" but "Self-Regulation at 3 creates a specific vulnerability under pressure: reactive decisions that damage relationships when it matters most."

Cite the scores
Every claim names the actual scores behind it.

Name the interaction type
The reader should be able to look it up in Chapter 5.

Strengths and risks in balance
Every trait pole has value; no profile is described as simply good or bad.

Tendencies, not certainties
Reports say "tends to" rather than "is", reflecting Whole Trait Theory.

Care with wellbeing
Wellbeing content is supportive, private and never framed as a performance judgement.

Three findings from outside personality psychology shape these rules. Fluent, easy-to-read material is judged more pleasing and more likely to be true, independent of its accuracy (Reber, Schwarz and Winkielman, 2004). Readers absorbed in a narrative counter-argue less and accept its implications more readily (Green and Appel, 2024). And people readily accept vague, generally true descriptions of themselves, the Barnum effect (Rust, Kosinski and Stillwell, 2020). A well-written interaction narrative is therefore more persuasive than its evidence may justify.

The engine adds four safeguards:

Show uncertainty	Scores are reported with their measurement band; profiles within one STEN point of a threshold are marked as borderline.
Separate evidence from interpretation	Each claim shows the scores and interaction type behind it, visibly distinct from the narrative.
State what the profile does not show	Every report includes a short section on its limits.
Advise process, not outcome	Development advice describes concrete steps in concrete situations. Imagining the steps toward a goal improves progress; imagining the goal achieved does not (Taylor, Pham, Rivkin and Armor, 1998).

Register for different readers

EMPLOYEES

Employees reading their own reports are addressed directly, in the second person, because identification with a narrative increases engagement with development advice.

MANAGERS

Managers reading reports about others receive a more neutral, third-person register, which reduces the risk of a persuasive story overriding their own observations of the person.

Quality checks

Before release, each report is tested against five criteria:

- ? **Could this report have been written about anyone else?** If yes, it fails.
- ? Is every claim traceable to a score and an interaction?
- ? Are flags placed first and handled correctly?
- ? Is the language behavioural and specific rather than generic?
- ? Are the recommendations matched to the assessment purpose?

The first test is a direct defence against the Barnum effect. A report that would feel accurate to almost anyone is not telling the reader anything about this particular person.

Traceability

Because every statement passes through the knowledge base, each sentence can be linked to its source: a trait score, a Tier 1 specification or Tier 2 rule, and a protocol step. This makes the report auditable. A practitioner who disagrees with a statement can see exactly which rule produced it and challenge the rule rather than the engine. Over time, challenges of this kind are the main route by which the knowledge base improves.

The engine version is recorded with every report. After any change to the engine or the knowledge base, a fixed set of test profiles is re-run and compared with earlier outputs, so that changes in reports can be explained.

Evaluating the engine

Three properties need empirical testing.

CONSISTENCY
Does the same profile produce materially the same report on repeated runs?

FIDELITY
Do report statements match what the knowledge base specifies?

EXPERT AGREEMENT
Do qualified psychologists, blind to the source, rate engine reports as accurate and useful as reports written by experts?

A fourth check concerns fairness: reports generated for identical scores but different demographic details should not differ in language or tone.

RESEARCH IN PROGRESS

For a sample of profiles we will report agreement between repeated runs, the percentage of statements traceable to the knowledge base, the number of statements contradicting it, and mean expert ratings of engine reports compared with expert-written reports. User satisfaction will not be used as evidence of accuracy, for the reasons given under Language rules.

What the system does not claim

THE BRAIN does not claim that the engine understands people, that it can replace a psychologist, or that it produces new psychological theory. It claims something narrower: that a carefully built knowledge base can be applied consistently and at scale when the reasoning engine is tightly constrained. **Final decisions about people remain with trained humans.**

CHAPTER 7 REFERENCES

Green, M. C., & Appel, M. (2024). Narrative transportation. *Advances in Experimental Social Psychology*, 70, 1–82.

Reber, R., Schwarz, N., & Winkielman, P. (2004). Processing fluency and aesthetic pleasure. *Personality and Social Psychology Review*, 8(4), 364–382.

Rust, J., Kosinski, M., & Stillwell, D. (2020). *Modern psychometrics* (4th ed.). Routledge.

Taylor, S. E., Pham, L. B., Rivkin, I. D., & Armor, D. A. (1998). Harnessing the imagination. *American Psychologist*, 53(4), 429–439.

PART III

From Individuals to Organizations

Chapter 8 Archetypes: From Profiles to Patterns

Chapter 9 Corporate Brain: Linking Behaviour to Business Strategy

CHAPTER 8

Archetypes: From Profiles to Patterns

SUMMARY This chapter describes how THE BRAIN summarizes Big Five profiles as archetypes that managers can use. It defines 32 patterns from high and low combinations of the five domains, groups them into 10 core archetypes and 22 family variants with neutral names, adds 26 behavioural modifiers, and links facets to observable workplace signals.

Why archetypes, and their risks

Managers cannot hold 30 facet scores in mind. They can remember a short, meaningful pattern. Archetypes trade precision for usability. That trade has a known cost, discussed in Chapter 3: categories can hide differences between people with similar labels and exaggerate differences between people near a boundary. THE BRAIN therefore uses archetypes as entry points to the full profile, never as replacements for it. Every archetype is reported with its underlying scores and live interactions, and with an indicator of how close the profile sits to an archetype boundary.

Thirty-two patterns

Each of the five domains is classified as High (H) or Low (L), giving 2 to the power of 5, or 32, patterns. Patterns are written in the order O–C–E–A–S, where S is Emotional Stability. Ten patterns form the core archetypes.

READING A PATTERN CODE

O	C	E	A	S
Openness	Conscientiousnes	Extraversion	Agreeableness	Emotional Stability

Each position is H or L. For example, **H-H-L-L-H** is the Builder: high Openness, high Conscientiousness, lower Extraversion, lower Agreeableness, high Emotional Stability.

32	10	22	26
patterns	core archetypes	family variants	modifiers

CODE	ARCHETYPE	PATTERN	CORE BEHAVIOUR	BEST ENVIRONMENT
A01	**Elite**	H-H-H-H-H	Energetic, capable, disciplined, socially effective and stable; turns ideas into execution	High-ownership, challenging roles
A02	**Operator**	L-H-H-H-H	Structured and execution-oriented; turns plans into consistent delivery	Clear objectives, measurable outcomes
A03	**Integrator**	H-H-L-H-H	Thinks and executes; coordinates people, information and priorities	Cross-functional, collaborative work
A04	**Dominant**	H-H-H-L-H	Assertive, competitive, results-driven; pushes decisions and performance	Competitive, high-accountability settings
A05	**Builder**	H-H-L-L-H	Independent deep worker; turns complex problems into tangible solutions	Independent, technically demanding work
A06	**Visionary**	H-L-H-H-H	Generates ideas and opportunities; sees future directions	Ambiguous, exploratory, innovation-focused
A07	**Rainmaker**	H-L-H-L-H	Energetic, persuasive, commercially driven; pursues opportunities	External, target-driven roles
A08	**Creator**	H-L-L-H-H	Quietly creative; generates novel ideas but may not push them forward	Low-politics, time to think
A09	**Supporter**	L-H-L-H-H	Reliable, cooperative, stable; strengthens team functioning	Stable teams with supportive leadership
A10	**Executor**	L-H-L-L-H	Quiet, dependable, task-focused	Clearly defined deliverables

Table 8.1 Core archetypes (from the Corporate Brain Master Sheet).

Reading the core archetypes

All ten core archetypes share high Emotional Stability. They describe how stable people differ in where they direct their energy. The Elite profile combines every adaptive pole and is, for that reason, both valuable and easily overloaded: organizations tend to give such people everything. The Operator and Executor deliver consistently and are often under-recognized because their reliability makes them invisible. The Visionary, Rainmaker and Creator share high Openness and lower Conscientiousness; they generate possibilities and benefit from partners who convert ideas into plans. The Integrator and Supporter hold teams together; the Dominant and Builder push work forward in different ways, one through people and one through independent effort.

Explore
Visionary · Rainmaker · Creator

Deliver
Operator · Executor · Dominant · Builder

Connect
Integrator · Supporter

A practical rule follows. Most strong teams need a mix: some people who explore, some who deliver, and some who connect. Chapter 9 and Part V, Section G, develop this idea.

Family variants and neutral naming

The remaining 22 patterns are family variants. Each is mapped to a primary and a secondary core archetype plus one or two modifiers. Earlier internal versions used labels such as "Broken" or "Dysfunctional". Labels like these can harm the people they describe, invite misuse in hiring, and would not pass ethical review. All 22 have been renamed to describe behaviour rather than judge the person.

H-H-H-H-L	Overextended Achiever	L-H-H-H-L	Pressured Operator
H-H-H-L-L	Intense Driver	L-H-H-L-H	Standards Keeper
H-H-L-H-L	Attuned Collaborator	L-H-H-L-L	Assertive Executor
H-H-L-L-L	Independent Specialist	L-H-L-H-L	Pressured Supporter
H-L-H-H-L	Variable Energizer	L-H-L-L-L	Reserved Executor
H-L-H-L-L	Reactive Challenger	L-L-H-H-H	Social Connector
H-L-L-H-L	Cautious Collaborator	L-L-H-H-L	Expressive Connector
H-L-L-L-H	Reflective Thinker	L-L-H-L-H	Bold Challenger
H-L-L-L-L	Low-Structure Explorer	L-L-H-L-L	Spontaneous Mobilizer
L-L-L-H-H	Steady Helper	L-L-L-H-L	Support-Seeking Profile
L-L-L-L-H	Steady Observer	L-L-L-L-L	Low-Energy Profile

Table 8.2 Family variants with neutral labels (pattern O-C-E-A-S). All platform reports, workbooks and website materials use these labels.

A NOTE ON LOWER EMOTIONAL STABILITY

Many family variants share lower Emotional Stability. These profiles call for particular care. A lower score on Emotional Stability describes greater sensitivity to stress, not weaker capability, and several of these profiles, such as the Overextended Achiever and Attuned Collaborator, combine high capability with high sensitivity. The right response is usually better support and workload design, not lower expectations.

Modifiers

Archetypes describe stable patterns. Modifiers describe current conditions that change how a manager should respond. THE BRAIN defines 26 modifiers, each with a trigger, a data source, its meaning, and adjustments to management, communication, autonomy and feedback. Examples:

MODIFIER	TRIGGER	MANAGEMENT ADJUSTMENT
Overcommitment Risk	Accepts excessive work, many parallel priorities, missed commitments	Reduce competing priorities; review commitments regularly
Emotional Reactivity	Strong reactions to criticism, conflict or pressure	Slow escalation; give feedback privately and specifically
Conflict Risk	Repeated interpersonal friction	Clarify decision rights; give behavioural feedback with examples
High Autonomy Need	Resists close supervision; performs best with ownership	Agree outcomes, not methods
High Structure Need	Performs best with clear process and expectations	Provide explicit plans and checkpoints

Table 8.3 Selected modifiers. The full approach is described in Appendix D.

Several modifiers draw on workload and behavioural data rather than questionnaires. This links archetypes to the signals described next. Modifiers matter because they are the part of the profile that changes fastest and that managers can most directly influence. **Many modifier adjustments change the situation, such as priorities, decision rights and feedback timing, rather than asking the person to change.**

From facets to observable signals

THE BRAIN maps each Big Five facet to everyday workplace behaviours, the contexts where they appear, the high and low signals, and the data source that could capture them. Thirty facet-level metrics come from tools organizations already use: 27 from project management systems, and 3 from HRMS, calendar and communication tools.

FACET	EXAMPLE METRIC	SOURCE
O1 Fantasy	Number of tasks created with multiple subtasks	Project management
O2 Aesthetics	Share of tasks with image or design attachments	Project management
O3 Feelings	One-to-one meetings attended per month	HRMS / calendar
C2 Order	Share of tasks with complete fields and due dates	Project management
C5 Self-Discipline	Share of tasks completed by due date	Project management

Table 8.4 Example facet metrics.

These metrics are hypotheses.

Each must show a meaningful correlation with the questionnaire facet it is meant to reflect, within the same role, before it can be used. Many metrics reflect job design as much as personality: a project manager creates more subtasks than an engineer regardless of Openness.

RESEARCH IN PROGRESS

For each of the 30 metrics we will report its correlation with its target facet score, the sample size and a verdict: retain, revise or drop.

Retain	Revise	Drop

Behavioural data from work systems is collected only with consent, used in aggregate wherever possible, and governed by the safeguards in Chapter 10.

CHAPTER 8 REFERENCES

Kosinski, M., Stillwell, D., & Graepel, T. (2013). *PNAS*, 110(15), 5802–5805.

Rust, J., Kosinski, M., & Stillwell, D. (2020). *Modern psychometrics* (4th ed.). Routledge.

CHAPTER 9

Corporate Brain: Linking Behaviour to Business Strategy

SUMMARY This chapter presents Corporate Brain, a behavioural analytics platform that implements THE BRAIN, as a case of applied workforce intelligence. It describes the platform's three-level structure, the mapping of six organizational goals across five functions, and a library of 300 strategies that link goals, functions and archetypes to concrete management actions. It closes with a framework for case studies and a first pilot case.

The platform

Corporate Brain combines psychometric assessment, technical assessment and project data in one system. It serves three levels of user:

Leadership
Organization-wide patterns, risks and alignment with strategic goals.

Managers
Team composition, individual archetypes, live interactions and management guidance.

Employees
Their own results and development guidance. Every employee can see their own report.

Each client organization's data is held in a separate database. The platform is designed for compliance with India's Digital Personal Data Protection Act and for GDPR readiness, and has undergone vulnerability and penetration testing (Chapter 10).

From goals to people

Most assessment platforms stop at the individual. Corporate Brain starts from what the organization is trying to achieve. It defines six goals, each with a strategic question, sub-goals, KPIs, and leading and lagging indicators.

6 goals
what the organization wants
→
× 5 functions
30 goal–function cells
→
× 10 archetypes
300 strategies

GOAL	STRATEGIC QUESTION	PEOPLE CONTRIBUTION (EXAMPLES)
G1 Financial	How do we make the business financially stronger?	Revenue quality, pricing discipline, utilization, cost discipline
G2 Growth	How do we become bigger?	Generating opportunities, acquiring customers, developing markets
G3 Customer	Are customers choosing us, succeeding with us and staying?	Understanding needs, building trust, solving problems
G4 Operational	How do we deliver reliably, efficiently and at quality?	Process discipline, planning, quality ownership, continuous improvement
G5 Innovation	How do we create what comes next?	Generating ideas, experimenting, turning ideas into products and processes
G6 Capability	Do we have the people and skills the strategy needs?	Learning, developing others, succession, retaining key talent

Table 9.1 Organizational goals (Corporate Brain Master Sheet).

Each goal also lists its key tensions, such as growth versus profitability, speed versus control, or experimentation versus efficiency. These tensions matter because different archetypes resolve them differently. A team of Rainmakers will resolve the growth-versus-profitability tension in favour of growth; a team of Operators will resolve it in favour of control. Knowing which way a team leans is the first step to balancing it.

Goals across functions

Each goal is broken down for five functions: Marketing, Sales, IT, HR and Leadership. That gives 30 goal–function combinations. Each combination specifies a primary objective, sub-goals, KPIs, the people outputs required, the key competencies and the technical skills to assess.

EXAMPLE · FINANCIAL GOAL × SALES

Primary objective	Maximize profitable revenue, not revenue alone.
KPIs	Gross margin, win rate, forecast accuracy, discount levels.
Required outputs	Closing profitable deals, protecting pricing, prioritizing high-value opportunities.
Key competencies	Negotiation, commercial judgment, relationship management.

EXAMPLE · CAPABILITY GOAL × HR

Primary objective	Ensure the organization has the skills its strategy requires.
KPIs	Internal fill rate for key roles, time to competence, retention of high performers.
Required outputs	Accurate skills mapping, targeted development, credible succession plans.
Key competencies	Talent judgment, coaching, workforce planning.

The strategy library

The strategy library crosses goals, functions and archetypes. For each combination it gives a strategic objective, the archetype's best contribution, how to allocate work, a manager strategy and a communication style. The library holds 300 entries: 30 goal–function cells multiplied by 10 core archetypes.

GOAL × FUNCTION	ARCHETYPE	BEST CONTRIBUTION	MANAGER STRATEGY	COMMUNICATION
Financial × Marketing	Elite	Connect strategy, experimentation and execution	Give an ambitious financial outcome and authority to choose the route	Strategic, direct, outcome-focused
Financial × Marketing	Operator	Process optimization and disciplined execution	Set clear financial targets and operating thresholds	Structured, numerical, specific
Financial × Marketing	Integrator	Cross-functional coordination	Give ownership of cross-functional initiatives	Collaborative, contextual

Table 9.2 Strategy library excerpt.

A parallel Big Five strategy map, at the level of single traits, gives function-level guidance. In Marketing, for example, Openness drives positioning and differentiation; low Openness in campaign roles risks generic messaging. In Talent Management, Agreeableness supports trust and conflict resolution.

A NOTE ON EVIDENCE

The strategies are expert-written professional guidance. They have not yet been tested as interventions, and Chapter 11 proposes how to evaluate them.

How it works in practice

ILLUSTRATIVE COMPOSITE

A mid-sized financial-services firm sets a Financial goal for the year: improve margin while holding revenue. Its sales team of twelve is assessed. The platform shows that most of the team are Rainmakers and Dominants, strong at winning business but likely to trade margin for volume. Two members show the Ambition × Self-Regulation Conflict pattern. The strategy library recommends pairing Rainmakers with an Operator on deal review, setting margin-based rather than volume-based targets, and giving the two Conflict-pattern employees coaching on decisions under pressure. Wellbeing information plays no part in these recommendations.

This example is a composite for illustration. It does not describe a real organization.

Case studies

Corporate Brain has been deployed in organizations across automotive, banking, insurance, broking, IT services and urban mobility. Anonymized cases will be reported using a common structure.

ELEMENT	CONTENT REQUIRED
Context	Industry, organization size band, function assessed, business goal
Sample	Number of employees assessed; demographic summary
Findings	Archetype distribution, main live interactions, flags raised (excluding wellbeing details)
Actions taken	What the organization changed
Outcomes	Before-and-after measures on agreed KPIs, with time frame
Client perspective	A short comment attributed by role, with consent
Limitations	Confounds, sample size, absence of a comparison group

Table 9.3 Case study template.

CASE 1 · PILOT DEPLOYMENT

A sales team in an Indian enterprise

Context. A privately held Indian enterprise deployed Corporate Brain in October 2026 for its sales function. The report was generated on 30 October 2026 for the organization's leadership. The organization is not named.

Sample. Seven of eight employees in the sales team completed the assessment (88 per cent coverage). All seven sit in one team, so team and company figures are identical.

Trait profile. Scores are on Corporate Brain's 0 to 100 reporting scale.

TRAIT	TEAM AVERAGE (0–100)	LOWER	MIDDLE	HIGHER
Openness	62	2	0	5
Conscientiousness	73	0	2	5
Extraversion	53	3	2	2
Agreeableness	64	0	3	4
Emotional Stability	55	1	4	2

Table 9.4 Case 1 trait distribution (n = 7).

The team is strongest on Conscientiousness and lightest on Extraversion. No member scored in the lower band on Conscientiousness or Agreeableness. Extraversion was the most spread out, with three members in the lower band.

ARCHETYPE	PATTERN (O-C-E-A-S)	MEMBERS	SHARE
Builder	H-H-L-L-H	2	29%
Integrator	H-H-L-H-H	2	29%
Visionary	H-L-H-H-H	1	14%
Dominant	H-H-H-L-H	1	14%
Supporter	L-H-L-H-H	1	14%

Table 9.5 Case 1 archetype distribution.

Five of the ten core archetypes were present. Not represented: Elite, Operator, Rainmaker, Creator and Executor.

Functional balance. The platform grouped archetypes into four contribution areas:



Ideas and opportunities · leading and pushing · delivery and process · teamwork and coordination

Findings

A delivery-strong, coordination-heavy team with thin commercial drive. For a sales team, the low share of opportunity-generating (Visionary) and decision-pushing (Dominant) profiles is notable. No Rainmaker, the archetype most associated with pursuing external opportunities, was present. Growth may depend on one or two people.

Low Extraversion in a client-facing function. An average of 53, with three of seven in the lower band, is a possible mismatch with sales work. Barrick and Mount (1991) found Extraversion predicts performance in sales; this is a hypothesis to examine against the team's sales results, not a conclusion.

Harmony may suppress challenge. High Agreeableness alongside few challenging profiles fits the risk the platform flagged: hard conversations may be avoided. This resembles the Suppression pattern at team level (Chapter 5) and the complementary cognition argument for mixing profiles (Part V, Section G).

A pressure pattern in two members. Two members showed strong capability with lower Emotional Stability, the Overextended Achiever pattern. The platform framed this as a conversation to have, not a label, in line with the wellbeing safeguards in Chapter 10. Individual details are not reported here.

Recommendations generated. The platform recommended developing Extraversion-related behaviours through training and work design rather than expecting them to appear, and assessing the remaining employee, since with small numbers one person can shift the picture.

Limitations. With seven people, every percentage is one or two individuals, and no statistical inference is possible. The case shows how the framework reads a real team and what it recommends; it does not test whether those readings are correct. Its value will grow once outcomes are recorded and the case is joined by larger deployments.

RESEARCH IN PROGRESS

Further anonymized cases from banking or automotive, insurance and IT services will follow, each reporting actions taken and before-and-after KPIs over at least one quarter.

What the cases can and cannot show

Case studies show how a framework is used and what changes follow. They cannot prove that the framework caused those changes, because many other things change at the same time. Stronger evidence would come from comparing teams that used the platform's recommendations with similar teams that did not. Chapter 11 proposes such a study.

CASES CAN SHOW

How a framework is used and what changes follow.

CASES CANNOT SHOW

That the framework caused those changes. That needs a comparison with similar teams that did not use it.

PART IV

Responsibility and the Future

Chapter 10 Validity, Ethics and Governance

Chapter 11 Future Directions and Conclusion

CHAPTER 10

Validity, Ethics and Governance

SUMMARY This chapter sets out the evidence THE BRAIN must provide to be trusted, and the safeguards that govern its use. It covers reliability, validity and norms, fairness in hiring, the handling of wellbeing information, consent for behavioural data, and compliance with Indian and European data protection law.

The evidence a framework owes its users

The Standards for Educational and Psychological Testing (AERA, APA and NCME, 2014) set out what any assessment used for decisions must show. In plain terms:

Reliability
It must measure consistently

Validity
It must measure what it claims and predict what matters

Norms
It must be interpreted against a relevant comparison group

Fairness
It must work fairly across groups

This chapter sets out THE BRAIN's programme of evidence against each standard. We publish the programme before the results because organizations deserve to know what to ask any assessment provider, including us.

Reliability

Reliability asks whether the instrument gives consistent results. Two forms matter most. Internal consistency asks whether the items measuring a trait agree with each other. Test–retest reliability asks whether a person receives similar scores when assessed again a few weeks later.

EVIDENCE	WHAT WILL BE REPORTED	TARGET
Internal consistency	Cronbach's alpha and McDonald's omega for each domain and facet	0.70 or above for domains; facets below 0.60 flagged for revision
Test–retest	Correlation over a 2 to 6 week interval, at least 100 participants	0.70 or above

Table 10.1 Reliability evidence programme.

Validity

Structural validity asks whether the data support the intended five-domain structure, tested by confirmatory factor analysis. Convergent validity asks whether scores agree with an established Big Five measure such as the BFI-2 or IPIP-NEO, with a target correlation of 0.60 or above for matching domains. Criterion validity asks whether scores predict work outcomes such as manager-rated performance. Incremental validity of interactions is covered in Chapter 5 (Testing the taxonomy).

VALIDITY TYPE	QUESTION	METHOD
Structural	Do responses form the intended five domains?	Confirmatory factor analysis (CFI, TLI, RMSEA, SRMR)
Convergent	Do scores match an established measure?	Correlation with BFI-2 or IPIP-NEO in a joint sample
Criterion	Do scores predict work outcomes?	Correlation with manager ratings and other outcomes
Incremental	Do interactions add prediction?	Pre-registered, split-sample tests (Chapter 5)

Table 10.2 Validity evidence programme.

Norms for Indian working populations

Most Big Five norms come from Western samples. Norms for Indian professionals are needed for STEN scores to be meaningful: a STEN score says how someone compares with a reference group, and the wrong reference group produces misleading comparisons. Norm samples will be described by size, sector, function, seniority, gender, age and region, with domain means and standard deviations reported by subgroup where samples allow.

Fairness in hiring

Personality measures typically show smaller group differences than cognitive tests, but fairness must be shown, not assumed. Use rules in hiring:

- ✓ Assessment results inform structured interviews; they never decide alone.
- ✓ No candidate is rejected automatically by a score or flag.
- ✓ Only traits with a demonstrated link to the role are weighted.
- ✓ Wellbeing and distress indicators are excluded from hiring entirely (see Wellbeing information).
- ✓ Learning Style traits are treated as preferences and are not weighted in hiring.
- ✓ Candidates are told what is assessed and why, and can request their results.

Adverse impact is assessed with the four-fifths rule: if the selection rate for any group is below 80 per cent of the rate for the highest-scoring group, the trait's use in screening is reviewed (Rust, Kosinski and Stillwell, 2020). Because hiring-mode responses are inflated on desirable traits, separate norms are kept for hiring and development contexts.

Wellbeing information

THE BRAIN includes items related to wellbeing, including depression-related items, because organizations have a duty of care and because masked distress is a real organizational risk (Chapter 5). Such information is also among the most sensitive data a person can disclose. Using it to screen candidates could amount to discrimination on grounds of disability or health, including under India's Rights of Persons with Disabilities Act, 2016.

The framework therefore applies four rules.

No hiring use
Wellbeing indicators are not scored for decision purposes in recruitment and are never shown to hiring managers.

Individual first
In development settings, results go first to the individual.

Consent for referral
A wellbeing flag leads to an offer of confidential support, such as an employee assistance programme. Information is shared with a designated wellbeing contact only with the individual's consent.

Not a diagnosis
Reports state clearly that the indicators are not a clinical assessment.

Behavioural and communication data

Corporate Brain can analyze consented work emails and chats for communication style. This is powerful and intrusive. The following safeguards apply:

- Explicit, specific and revocable employee consent.
- Analysis of style features, not storage of message content beyond processing needs.
- Retention of 30 days, after which raw data is deleted.
- Employees can see their own results.
- No covert monitoring and no use for disciplinary purposes.

PROHIBITED INFERENCES

Digital records can predict sexual orientation, ethnicity and political affiliation with high accuracy (Kosinski, Stillwell and Graepel, 2013), and personality itself relates to political ideology (Gerber and others, 2011). Corporate Brain therefore never infers, stores or reports religion, caste, ethnicity, political views, sexual orientation, health conditions or other protected characteristics. Models are restricted to the work-relevant traits they were validated for, and are audited for proxies.

Dated profiles. Work changes personality over time: in a 20-year study, sustained high workload increased Openness, Extraversion and Agreeableness (Holman and Hughes, 2021). Every report carries its assessment date, and profiles older than 24 months are flagged for reassessment.

Data protection

The platform is designed to comply with India's Digital Personal Data Protection Act, 2023, and to be ready for the EU General Data Protection Regulation. Key measures include:

- A separate database for each client
- Purpose limitation
- Data minimisation
- The right to access and erase data
- Regular vulnerability and penetration testing

Ethical review for research use

Any use of platform data for the research described in this monograph requires ethical clearance and client agreement covering secondary research use. Data used for research is anonymized and pooled; no individual or organization is identifiable in published results.

CHAPTER 10 REFERENCES

American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (2014). *Standards for educational and psychological testing*. AERA.

Gerber, A. S., Huber, G. A., Doherty, D., & Dowling, C. M. (2011). The Big Five personality traits in the political arena. *Annual Review of Political Science*, 14, 265–287.

Government of India. (2016). The Rights of Persons with Disabilities Act.

Government of India. (2023). The Digital Personal Data Protection Act.

Rust, J., Kosinski, M., & Stillwell, D. (2020). *Modern psychometrics* (4th ed.). Routledge.

Soto, C. J., & John, O. P. (2017). The next Big Five Inventory (BFI-2). *Journal of Personality and Social Psychology*, 113(1), 117–143.

CHAPTER 11

Future Directions and Conclusion

SUMMARY This closing chapter looks ahead to new data sources and new forms of behavioural intelligence, proposes a research agenda for testing THE BRAIN, and summarizes the contributions and recommendations of this monograph.

New ways of measuring

Questionnaires depend on self-report, which people can shape to look good, especially when a job is at stake. Two developments may reduce this problem, and a third offers a different route.

Game-based assessments

Infer traits from choices and behaviour inside designed tasks. They are harder to fake and more engaging, though their validity evidence is still developing. Brain Interacts is exploring them as a complement to questionnaires.

Passive behavioural signals

Signals from work systems, described in Chapter 8, can show how people actually behave over time rather than how they describe themselves. Their promise depends entirely on consent, transparency and validation.

Adaptive testing

Based on item response theory, an adaptive test chooses each next question based on earlier answers, measuring each trait precisely with fewer items. This could make measuring 91 traits practical without very long questionnaires.

From one-off assessment to continuous intelligence

Most organizations assess people at a few moments: hiring, promotion, a development programme. Whole Trait Theory suggests behaviour is better understood as a distribution that shifts with situations. Combining periodic assessment with consented behavioural signals could let organizations see when someone's pattern is changing, such as a Contextual Switch becoming frequent under sustained pressure, and respond with support rather than surprise.

A situation layer

Funder's personality triad (Part V, Section B) suggests that persons, situations and behaviours must be understood together. THE BRAIN models persons in detail and behaviours through narratives, but does not yet model situations systematically. A planned addition is a short classification of the situations each role involves, such as high pressure, conflict, ambiguity and public scrutiny. Interactions whose triggers occur in the role would then be given more weight than those whose triggers rarely occur.

Team-level interaction

THE BRAIN analyses interactions within one person. The theory of complementary cognition (Part V, Section G) suggests analysing interactions between people: whether a team's members compensate for, amplify or conflict with one another. The eleven interaction types could be applied at team level. A team Ceiling Effect, for example, would describe a team in which no member has strong Active Listening, so that the team's collective communication is capped regardless of its members' other strengths.

A research agenda

Test the eleven types
Estimate interaction effects for Tier 1 pairs on real outcomes (Chapter 5).

Validate the instrument
Complete reliability, structure, convergent validity and Indian norms (Chapter 10).

Validate behavioural metrics
Test each of the 30 facet metrics against questionnaire scores within roles (Chapter 8).

Evaluate the narrative engine
Measure consistency, fidelity, fairness and expert agreement (Chapter 7).

Test organizational impact
Compare teams that act on platform recommendations with matched teams that do not, on agreed KPIs over at least two quarters.

Study culture
Examine how interaction patterns differ across Indian regions, sectors and organizational cultures, and in later international deployments.

Contributions

This monograph has made four contributions.

Foundation	It has argued, on the evidence, that the Big Five is the most defensible foundation for organizational behavioural analytics.
Taxonomy	It has proposed an eleven-type taxonomy of trait interactions, giving practitioners a vocabulary for what combinations of traits do.
Architecture	It has described THE BRAIN, a tiered architecture that applies interaction reasoning across 91 traits, and a model of automated reasoning as a constrained, traceable instrument.
Application	It has shown how individual profiles can be linked to organizational goals through archetypes and a strategy library, implemented in Corporate Brain.

FOR LEADERS

Recommendations for leaders

- → Use the Big Five, not type-based tools, as the base for people decisions.
- → Read profiles as combinations, not lists of scores.
- → Start from business goals and work back to people.
- → Insist that any automated reasoning in assessment is constrained, explainable and checked by humans.
- → Keep wellbeing information for support, never selection.
- → Ask vendors, including us, for their validation evidence.

CLOSING

Personality science has given organizations good instruments for measuring people. The next step is to understand them. That requires looking at how traits work together, linking that understanding to what organizations are trying to achieve, and doing it with care for the people being assessed. THE BRAIN is offered as one step in that direction, open to testing and to improvement.

PART V

Evidence Review

A Foundations of the Big Five
B Persons, Situations and States
C The Interaction Question
D Outcomes Beyond the Job
E Digital Traces and Measurement
F How People Read and Use Reports
G Teams as Cognitive Systems
H Analysis of THE BRAIN's Own Materials
I Synthesis

INTRODUCTION

The Research Base of THE BRAIN

This review analyses the sources we assembled for this work: papers and books on the Big Five, interaction, measurement and report design, together with THE BRAIN's own internal materials, including the 4,095-pair interaction matrix. Its purpose is to give each chapter a documented evidence base and to show readers how we weigh evidence.

THE CONCLUSION THAT SHAPES THE MONOGRAPH

The strongest large-scale test of Big Five trait-by-trait interactions to date (Vize and others, 2022) found that such interactions are rare and small when tested as simple statistical products of broad traits. THE BRAIN's claims do not fall with that finding, but they must be framed around it. Section C explains how.

How the sources were analysed

Each source was read in full. For each, the review records what was studied and how, the main findings with their key figures, the limits of the evidence, and what the source implies for THE BRAIN and Corporate Brain. Where a source has methodological weaknesses, these are stated.

GRADE	MEANING	EXAMPLE
A	Large-sample meta-analysis, registered report or authoritative review	Barrick & Mount (1991); Vize and others (2022)
B	Peer-reviewed empirical study or established theory paper	Holman & Hughes (2021); Funder (2006)
C	Peer-reviewed study with notable limits, or a narrative review	Yu and others (2024); Campbell and others (2023)
D	Unreviewed or weakly reviewed work; context only	Sweeney (2023) dissertation

Table V.1 Evidence grades.

SOURCES

Map of sources to chapters

SOURCE	GRADE	MAIN USE
Costa & McCrae (1995), Domains and facets	A	Ch. 2, Ch. 6
Barrick & Mount (1991), Big Five and job performance	A	Ch. 4
Funder (2006), The personality triad	B	Ch. 3, Ch. 5
Jayawickreme, Zachry & Fleeson (2019), Whole Trait Theory	B	Ch. 2, Ch. 5, Ch. 11
Holman & Hughes (2021), Traits and job characteristics over 20 years	B	Ch. 4, Ch. 9
Vize and others (2022), Do the Big Five interact?	A	Ch. 3, Ch. 5, Ch. 10
Campbell, Exley & Doyle (2023), Big Five and financial planning	C	Ch. 4, Ch. 9
Yu, Chang & Li (2024), Personality, social support and wellbeing	C	Ch. 4, Ch. 10
Sweeney (2023), Personality, learning and GPA	D	Cautionary note only
Phuthong (2023), OCEAN and knowledge sharing	C	Ch. 4
Gerber and others (2011), Big Five in politics	B	Ch. 4, Ch. 10
Kosinski, Stillwell & Graepel (2013), Digital records	A	Ch. 4, Ch. 8, Ch. 10
Rust, Kosinski & Stillwell (2020), Modern Psychometrics	A	Ch. 10
Reber, Schwarz & Winkielman (2004), Processing fluency	B	Ch. 7
Green & Appel (2024), Narrative transportation	B	Ch. 7
Taylor and others (1998), Mental simulation	B	Ch. 7, Ch. 11
Taylor, Fernandes & Wraight (2021), Complementary Cognition	C	Ch. 9
THE BRAIN materials and workbooks	Internal	Ch. 5 to 9

Table V.2 Map of sources to chapters.

SECTION A

Foundations of the Big Five

Costa and McCrae (1995): Domains and facets

Costa, P. T., Jr., & McCrae, R. R. (1995). Domains and facets: Hierarchical personality assessment using the Revised NEO Personality Inventory. *Journal of Personality Assessment*, 64(1), 21–50.

A GRADE

What it addresses. By the early 1990s the Five-Factor Model had become the dominant framework in personality psychology, but most attention went to the five broad factors. Critics argued that the factors were too broad for detailed assessment and that narrower traits predicted specific behaviour better. They did not, however, agree on which narrow traits to measure. Cattell proposed 16 factors; Jackson's Personality Research Form measured 20 of Murray's needs. None of these sets was built as a systematic division of the five-factor space. Costa and McCrae's answer was the domain-and-facet design of the NEO-PI-R: a 240-item questionnaire measuring each of the five domains through six facets, giving 30 facet scales.

Key arguments and evidence.

Personality is hierarchical. Specific behaviours cluster into habits, habits into narrow traits (facets), and facets into broad domains. Domains give a rapid overview; facets give a detailed picture.

Facets were chosen for conceptual and practical reasons. Each facet had to be a distinct aspect of its domain, represented in the existing literature, and roughly equal in breadth to the other facets in the same domain.

The structure replicates. With minor variations, the facet structure was reproduced in college students, in older adults tested by computer, and in spouse and peer ratings on the observer form.

Facets show discriminant validity. Using expert-judged descriptions, the authors showed that facets within the same domain have distinct patterns of correlates.

Facet analysis. The authors proposed interpreting any external scale by the specific facets it correlates with. The Dominance and Love scales of Wiggins' interpersonal measure turned out to correspond to a disagreeable form of Extraversion and an extraverted form of Agreeableness. Facet analysis reveals these blends where domain-level analysis would hide them.

Profile interpretation should be hierarchical. Interpretation begins with the domains, then examines facets, paying attention to facets that diverge from their domain.

Limits. The NEO-PI-R is a proprietary instrument with norms built mainly on American samples. The facet choices are one reasonable division of the domains, not the only one; the BFI-2 uses 15 facets, and DeYoung's model uses 10 "aspects".

WHAT IT MEANS FOR THE BRAIN

- THE BRAIN's Big Five Trait DNA adopts the NEO facet structure: 5 domains and 30 facets, from O1 Fantasy to N6 Vulnerability. This gives the framework a well-validated skeleton. THE BRAIN's items are written by Brain Interacts.
- Costa and McCrae's facet analysis is a close ancestor of THE BRAIN's approach. Both hold that blends of traits reveal what single labels conceal. Facet analysis interprets external scales; THE BRAIN interprets combinations within one person's profile.
- Their advice to read divergent facets is a form of interaction reasoning within a domain. A person with high overall Conscientiousness but low C6 Deliberation is a specific case THE BRAIN's Trait DNA can describe.
- The 91-trait inventory extends well beyond the Big Five. Facet analysis offers the method for showing how the additional 86 traits relate to the five-factor space: correlate each additional trait with the 30 facets to show where it sits and what it adds. This analysis is part of the research programme.

Barrick and Mount (1991): The Big Five and job performance

Barrick, M. R., & Mount, M. K. (1991). *Personnel Psychology*, 44(1), 1–26.

A GRADE

What it addresses. Before 1991, reviews concluded that personality predicted job performance poorly. Barrick and Mount argued that this was partly because no shared taxonomy existed: studies used hundreds of different scales, so findings could not accumulate. Once the Five-Factor Model offered a common language, scales could be classified into the five dimensions and their validities pooled.

Method. The authors classified personality scales from 117 studies (162 samples, total N = 23,994) into the five dimensions. They examined three criteria, job proficiency, training proficiency and personnel data such as salary or tenure, across five occupational groups: professionals, police, managers, sales and skilled or semi-skilled workers. Validities were corrected for measurement error and range restriction.

DIMENSION	PATTERN OF VALIDITY
Conscientiousness	Valid for all criteria in all occupational groups; the only dimension with consistent validity
Extraversion	Valid for managers and sales; valid for training proficiency
Openness to Experience	Valid for training proficiency across occupations
Emotional Stability	Smaller and less consistent than expected
Agreeableness	Small validities in most groups

Table V.3 Findings of Barrick and Mount (1991).

Limits. The meta-analysis pooled scales designed for other models, so classification required judgement. Later meta-analyses found Emotional Stability to be a more general predictor than this first study suggested. Effect sizes, even for Conscientiousness, are moderate.

WHAT IT MEANS FOR THE BRAIN

- **Conscientiousness deserves high base weight** for most roles, as THE BRAIN already gives it.
- **Role matters.** Extraversion predicting performance in sales and management, but not elsewhere, is the evidential basis for role-specific weighting in the Consolidation Model and the goal-by-function mapping in Corporate Brain.
- **Training proficiency is a distinct criterion.** Openness and Extraversion predict learning in training, which supports Openness carrying a higher development weight than acquisition weight.
- **The common-language argument transfers.** Barrick and Mount showed that a shared taxonomy allows evidence to accumulate. THE BRAIN makes the same argument one level up: a shared taxonomy of interaction types could allow evidence about trait combinations to accumulate.

Secondary overviews

During our research we also reviewed several broad, unattributed overviews of the Big Five. These are useful as orientation but are not cited as evidence. Where an overview raised a claim worth using, we traced the original source and cited it instead. This is a general rule of the evidence base: **secondary summaries point the way; primary sources carry the weight.**

SECTION B

Persons, Situations and States

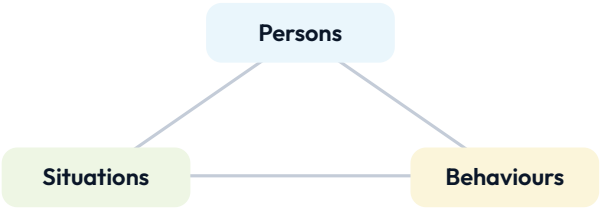
The three sources in this section answer a question every user of THE BRAIN will ask: if behaviour changes from one situation to the next, and people change over time, how much can a trait profile tell us? Together they support a clear answer. Traits describe stable tendencies, situations shift behaviour around those tendencies, and work itself slowly reshapes the tendencies.

Funder (2006): The personality triad

Funder, D. C. (2006). Towards a resolution of the personality triad: Persons, situations, and behaviors. *Journal of Research in Personality*, 40(1), 21–34.

B GRADE

What it addresses. The person–situation debate asked whether behaviour is driven mainly by stable personal characteristics or by the situation. By the 1980s nearly everyone called themselves an interactionist, yet the argument persisted. Funder argues the debate rests on a false choice and proposes the personality triad: persons, situations and behaviours, each understood and predicted in terms of the other two.



Evidence. Funder summarizes studies in which the same participants were videotaped in two social interactions with different partners in two laboratory sessions. Four trained assistants independently rated each tape on the Riverside Behavioral Q-sort, a set of 62 descriptions of observable social behaviour. Two findings appeared at once.

Situations changed behaviour

At the second session, participants were on average more relaxed, socially skilled, expressive and fluent. A small change had large, observable effects.

Individuals stayed consistent

Despite these average shifts, people kept their rank order on many behaviours. Someone more nervous than most at the first session was probably still more nervous than most at the second.

Funder also found that participants' own sense of how similar the two situations felt predicted how consistent their behaviour was. Psychological similarity, not objective features, predicted consistency.

Limits. The data are from laboratory interactions among students, a narrower range of situations than workplaces offer. The paper is a theoretical synthesis rather than a new study.

WHAT IT MEANS FOR THE BRAIN

- **A defence of trait-based assessment.** People do behave differently in different situations, but their relative standing is stable enough to predict from.
- **A warrant for the Contextual Switch type.** Situational triggers can produce large shifts while individual differences persist, which is exactly the pattern the type describes.
- **The missing third element.** Funder's triad suggests adding a situation layer to THE BRAIN, as proposed in Chapter 11.
- **Situational interventions.** If the behaviourally effective aspects of a situation can be identified, the situation can be changed. This supports Corporate Brain's modifiers, which often adjust priorities, decision rights or feedback timing rather than the person.

Jayawickreme, Zachry and Fleeson (2019): Whole Trait Theory

Jayawickreme, E., Zachry, C. E., & Fleeson, W. (2019). Whole Trait Theory: An integrative approach to examining personality structure and process. *Personality and Individual Differences*.

B GRADE

What it addresses. Trait theory describes stable individual differences. Social-cognitive theory explains behaviour through how people interpret situations, their expectations, goals and self-regulation. The two traditions advanced largely separately. Whole Trait Theory joins them.

Key ideas.

Traits have two parts. The descriptive part is a distribution of personality states: how extraverted or conscientious a person actually is from moment to moment. The explanatory part is the social-cognitive system that produces those states in particular situations.

Within-person variation is large and meaningful. Experience-sampling research shows that people's behaviour varies within themselves about as much as people differ from one another (Fleeson, 2001). Yet the centre of each person's distribution is stable, which is why trait scores predict.

Distributions settle the person–situation debate. Variation reflects situations and mechanisms; the stable centre reflects the trait.

Application beyond the Big Five, including to character traits such as honesty.

Implications for change. If states are produced by interpretations, goals and habits, changing those mechanisms could shift the distribution over time.

Limits. The explanatory mechanisms are still being specified, and most supporting evidence comes from student samples.

WHAT IT MEANS FOR THE BRAIN

- A STEN score of 8 describes where a person's distribution is centred, not how they behave every hour. Reports say "tends to" rather than "is".
- Contextual Switch, Masking and Paradox can all be described as properties of state distributions, giving the taxonomy a stronger theoretical anchor.
- Corporate Brain's consented behavioural signals are, in effect, repeated state measurements, which supports combining periodic questionnaires with continuous signals.
- Development works by changing habits, interpretations and goals, which then shift typical behaviour.

Holman and Hughes (2021): Personality and job characteristics over 20 years

Holman, D. J., & Hughes, D. J. (2021). *Journal of Occupational and Organizational Psychology*, 94(3), 762–788.

B GRADE

What it addresses. Organizational research has focused on how personality shapes work. Holman and Hughes tested whether work characteristics change personality, whether personality changes work characteristics, or both.

Method. The study used the Midlife in the United States (MIDUS) survey, a national sample of adults surveyed three times: 1994–95, 2004–06 and 2013–14. The analysis sample was the 1,049 participants employed at all three waves. Two job characteristics were studied, job discretion and workload, using latent change score models.

DIRECTION	RESULT
Workload → personality	Supported Consistently high workload led to increases in Openness, Extraversion and Agreeableness
Job discretion → personality	Not supported
Personality → job characteristics	Largely not supported Neuroticism predicted later job discretion
Reciprocal effects	Not found

Table V.4 Findings of Holman and Hughes (2021).

Limits. Workload and discretion were self-reported with short measures. The sample is American and middle-aged. The positive effect of workload may depend on accompanying resources; burnout was not measured.

WHAT IT MEANS FOR THE BRAIN

- **Profiles are not permanent.** Assessments carry a date and should be renewed every 12 to 24 months.
- **Organizations shape the profiles they later measure.** Strategy recommendations should work with current tendencies, not fixed types.
- **Development is plausible.** Demanding work, appropriately resourced, can build traits, supporting the Compounding Development type.
- **A caution on workload.** The same demands that grow some people overwhelm others. The Overcommitment Risk modifier and the wellbeing safeguards are the counterweight.
- **"Negative" traits can have adaptive consequences,** supporting the principle that every trait pole has value.

SECTION C

The Interaction Question

This section is the most important in the review. The central claim of this work is that traits interact and that interactions explain behaviour that single traits miss. One source tests a version of that claim directly and largely fails to support it. We think the work is far stronger for raising that finding first and answering it.

Vize, Sharpe, Miller, Lynam and Soto (2022): Do the Big Five interact?

Vize, C. E., Sharpe, B. M., Miller, J. D., Lynam, D. R., & Soto, C. J. (2022). Do the Big Five personality traits interact to predict life outcomes? Systematically testing the prevalence, nature, and effect size of trait-by-trait moderation. *European Journal of Personality* (Registered Report).

A GRADE

What it addresses. Researchers have long proposed that traits moderate one another. But interaction effects are notoriously hard to detect reliably, and no one had estimated how often they occur. Earlier work outside personality was discouraging: a review of 30 years of management research found a median interaction effect size of f^2 = .002 across 261 studies (Aguinis and others, 2005), and an exhaustive study of personality-by-situation interactions across more than 556,000 tests found little support beyond chance (Sherman and Pashler, 2019).

Method. The authors used the Life Outcomes of Personality Replication (LOOPR) project data, with the Big Five measured by the BFI-2. They tested every pairwise interaction of the five domains (10 pairs) against 81 self-reported life outcomes, giving 750 tests, with at least 1,350 participants per outcome. Each sample was split in two so that any interaction found in one half could be checked in the other. The study was registered in advance and powered to detect an interaction explaining about 1 per cent of additional variance. Robustness checks included demographic controls, alternative regression models and out-of-sample predictive accuracy.

Tests run	**750**
Significant in at least one half, before corrections	**132**
Some replicable evidence after demographic controls	**62**
Strongly replicable after all robustness checks	**40 (5.33%)**
Typical size of replicable interactions	**About 1% of additional variance**
Improvement in out-of-sample prediction from adding interactions	**None observed**

Table V.5 Findings of Vize and others (2022).

Where interactions did replicate, a common form was buffering. For antisocial behaviour, high Agreeableness or high Conscientiousness weakened the link between the other trait and the outcome.

Strengths. A registered report with a very large sample, built-in replication, multiple robustness checks and an explicit smallest effect of interest. It is the best available estimate of the base rate of Big Five interactions.

Limits that follow from the design.

- Only the five broad domains were tested, not facets or narrower traits.
- Interactions were tested as linear products of continuous scores.
- Outcomes were self-reported life outcomes, not observed workplace behaviour.
- Personality was self-reported with one instrument; the reliability of a product term is lower than that of its components, which makes interactions harder to detect.

What the finding means for THE BRAIN

The honest reading is this: at the level of the five broad domains, tested as linear products against general life outcomes, Big Five interactions are rare and small. If THE BRAIN claimed that multiplying broad trait scores adds substantial predictive power, Vize and colleagues would directly contradict it.

THE BRAIN makes a different, narrower set of claims. Each difference is a reason the finding may not transfer, but none is yet a demonstration that it does not.

VIZE AND OTHERS TESTED	THE BRAIN PROPOSES	WHY THIS COULD MATTER	STATUS
5 broad domains, 10 pairs	91 narrower traits, 58 expert-chosen priority pairs	Broad domains average out narrow effects; pairs chosen for theoretical reasons may behave differently	**Untested**
Linear product of continuous scores	Threshold patterns (high 7–10, low 1–4)	Some patterns may occur only in one corner of the score space	**Untested; threshold splits reduce statistical power**
General self-reported life outcomes	Specific behaviours in specific situations	An interaction may matter for behaviour under pressure, not broad outcomes	**Untested**
Predictive variance	Interpretation and decision support	A small average effect can still change the right decision for a particular person	**Conceptual**

Table V.6 How THE BRAIN's claims differ from those tested.

Two results from Vize and colleagues also support parts of THE BRAIN. The buffering pattern they found is a form of what THE BRAIN calls Compensation. And their finding that interactions are small even when real is consistent with THE BRAIN's tiered design, which hand-specifies only 58 of 4,095 pairs on the assumption that most pairs carry little consequence.

How this shapes the work

A theoretical taxonomy. The eleven types are presented as a theoretical taxonomy for interpretation, not as established predictive effects.

Rigorous standards. Tests follow Vize and colleagues' standards: pre-registration, split-sample replication, a stated smallest effect of interest, and samples of roughly 1,000 to 1,500.

Distinctive targets. The tests target THE BRAIN's distinctive claims: specific Tier 1 pairs, narrow traits, threshold patterns and behaviour-specific outcomes rated by others.

Out-of-sample checks. Does a model with Tier 1 interactions predict held-out outcomes better than one with main effects only?

A careful value claim. The primary claim is that the taxonomy gives practitioners a consistent, explicit way to reason about combinations they already reason about informally and inconsistently. The predictive claim is a hypothesis to be tested.

SECTION D

Outcomes Beyond the Job

Five sources examine what the Big Five predicts outside core job performance. They vary widely in quality. Read together, they confirm the main patterns in Section A and show how much a study's methods limit what it can claim.

Campbell, Exley and Doyle (2023): The Big Five and financial planning

Financial Services Review. A narrative review aimed at financial advisors.

C GRADE

TRAIT	REPORTED RELATIONSHIP WITH FINANCIAL OUTCOMES
Conscientiousness	Higher income, net worth and financial literacy; disciplined saving; may take less investment risk than optimal
Extraversion	Higher earnings, but weaker conversion of income into wealth
Agreeableness	Lower income and net worth
Neuroticism	Lower income and wealth; greater risk avoidance
Openness	Mixed results

Table V.7 Findings summarized by Campbell, Exley and Doyle (2023).

The review proposes three client profiles:

Resilient	Under-controlled	Over-controlled
Disciplined and calm	Impulsive, risk-seeking	Anxious, overly cautious

Limits. A narrative rather than systematic review; two authors are linked to a commercial personality-based financial assessment firm; several key findings come from the authors' own earlier work.

What it means for THE BRAIN. It supports the Financial goal mapping in Corporate Brain: Conscientiousness relates to cost discipline; Extraversion relates to revenue generation but not necessarily to profitable outcomes, which parallels the Rainmaker archetype's tendency to win volume at the expense of margin. The Resilient, Under-controlled and Over-controlled typology offers an external check on THE BRAIN's archetypes: platform data should be examined to see whether these broader profiles emerge.

Yu, Chang and Li (2024): Personality, social support and wellbeing

Education Sciences, 14, 1050. An online survey of 848 Taiwanese university students using a short Big Five measure.

C GRADE

Findings. Personality together explained about 45 per cent of the variance in wellbeing. Conscientiousness showed the strongest association (r = .58). Social support carried the whole effect of Agreeableness on wellbeing and part of the effect of Extraversion. The indirect effects were statistically significant but small.

Limits. A single cross-sectional survey cannot establish cause. A convenience sample of students. Three items per trait.

What it means for THE BRAIN. Different routes to wellbeing for different traits: social connection for agreeable and extraverted employees, autonomy and structure for conscientious employees. Agreeableness affecting wellbeing only through social support resembles THE BRAIN's Dependency type, with a situation rather than a trait as the enabler. Wellbeing findings are used only in development and support, never in hiring.

Sweeney (2023): Personality, learning styles and grade point average

Doctoral dissertation, St. John's University. A survey of 91 students.

D GRADE

Conscientiousness was the only trait related to GPA. The sample is too small for the number of relationships tested, and some reported statistics are internally inconsistent. "Learning styles" is a widely disputed construct.

What it means for THE BRAIN. Its one robust-looking result is better supported by large meta-analyses (Poropat, 2009). It is a caution about Learning Style traits: the evidence that matching instruction to learning styles improves learning is weak (Pashler and others, 2008). THE BRAIN describes these traits as preferences, not validated predictors, and does not weight them in hiring.

Phuthong (2023): OCEAN traits and knowledge sharing

Journal of Human, Earth, and Future. A survey of 200 Thai social-commerce users.

C GRADE

Extraversion and Neuroticism predicted sharing product knowledge and opinions; the other traits did not. Measurement was weak, effect sizes were very small, and causal language was used for cross-sectional data.

What it means for THE BRAIN. The direction of the Extraversion finding fits wider evidence. The Neuroticism finding suggests some information sharing is emotionally driven, so high message volume should not automatically be read as engagement. Specific figures are not relied on.

Gerber, Huber, Doherty and Dowling (2011): The Big Five in politics

Annual Review of Political Science, 14, 265–287.

B GRADE

Openness is the most consistent predictor of liberal ideology; Conscientiousness predicts conservative ideology. Effects are in many cases comparable in size to standard demographic predictors. The authors note concerns about short personality measures and inconsistent findings across contexts.

What it means for THE BRAIN.

- The Big Five predicts outcomes far beyond work, supporting its standing as a general model.
- Measurement matters. Measuring 91 traits creates pressure toward short scales, which reduce reliability and weaken every downstream inference.
- Context changes effects, strengthening the case for Indian norms and local validation.
- Because personality data can predict political views, Corporate Brain never infers, stores or reports political orientation or other protected characteristics.

SECTION E

Digital Traces and Measurement Standards

Kosinski, Stillwell and Graepel (2013): Private traits from digital records

PNAS, 110(15), 5802–5805.

A GRADE

Method. More than 58,000 volunteers provided their Facebook Likes, demographic profiles and psychometric test results. The authors reduced the Likes data to its main dimensions and used regression models to predict each attribute, testing accuracy on people not used to build the model.

ATTRIBUTE PREDICTED FROM LIKES	ACCURACY REPORTED
Sexual orientation (men)	88% correct classification
African American vs Caucasian American	95%
Democrat vs Republican	85%
Age	r = .75
Openness	r = .43
Extraversion	r = .40

Table V.8 Findings of Kosinski, Stillwell and Graepel (2013).

Limits. Self-selected volunteers; prediction worked best for traits with visible preferences; individual-level accuracy is moderate.

What it means for THE BRAIN.

- Behaviour carries personality signal, which is the foundation for the 30 facet metrics.
- Accuracy varies by trait. Some facet metrics will work and others will fail, so each is validated separately.
- The same methods reveal what must stay private, so protected-characteristic inference is explicitly prohibited and audited.
- Consent is necessary but not sufficient. Data minimization, analysing style features only and deleting raw data after 30 days, is the stronger protection.

Rust, Kosinski and Stillwell (2020): Modern Psychometrics

Modern psychometrics: The science of psychological assessment (4th ed.). Routledge.

A GRADE

Content most relevant to THE BRAIN.

Questionnaire construction
Blueprint, item writing, piloting, item analysis, reliability and validity, then standardization.

Reliability
And the standard error of measurement, which converts reliability into a band of uncertainty around each score.

Validity
In its several forms.

Standardization
Against a relevant norm group.

Bias
Item bias, lack of measurement invariance and adverse impact, commonly flagged by the four-fifths rule.

Social desirability
Especially in high-stakes settings.

Forced-choice formats
Make faking harder but complicate comparisons between people.

The Barnum effect
Acceptance of a report is not evidence of its validity.

Ethics
As integral to psychometrics, not an add-on.

What it means for THE BRAIN.

- It is the template for Chapter 10.
- Report the standard error. A person scoring 7 may truly be a 6, so threshold-based interaction rules should use bands or flag near-boundary profiles.
- Adverse impact is tested with the four-fifths rule for every screening trait.
- Hiring-mode assessments will be inflated on desirable traits, which calls for separate norms and consideration of forced-choice items for key traits.
- The quality test "could this report have been written about anyone else?" is a direct defence against the Barnum effect.
- Adaptive testing could make measuring 91 traits practical.

SECTION F

How People Read and Use Reports

THE BRAIN's final output is a written narrative read by managers and employees. Three sources from outside personality psychology explain how people process such reports. **The qualities that make a report engaging, a clear story, smooth reading, vivid images of success, also make it more persuasive than its evidence may justify.**

Reber, Schwarz and Winkielman (2004): Processing fluency

Personality and Social Psychology Review, 8(4), 364–382.

B GRADE

The more easily people can process something, the more positively they respond to it. Features long associated with beauty, such as symmetry, clarity, repetition and familiarity, all make processing easier. Information that is easy to process also tends to be judged more likely to be true.

What it means for THE BRAIN. A well-designed report feels more accurate than it is. Reports should be readable but should also show uncertainty visibly. Evaluation must use outcomes and expert agreement, never user satisfaction alone.

Green and Appel (2024): Narrative transportation

Advances in Experimental Social Psychology, 70, 1–82.

B GRADE

Narrative transportation is an immersive state in which attention, imagery and feelings are absorbed in a story. Transported readers counter-argue less and are more likely to adopt beliefs implied by the story. First-person narratives produce more identification. Because stories reduce critical scrutiny, false information embedded in a narrative can be persuasive and hard to correct.

What it means for THE BRAIN. The narrative format is powerful, which is both its value and its risk. Reports separate evidence from interpretation, include a section on what the profile does not show, and invite the reader to test the narrative against observed behaviour. Employees' own reports use direct address; manager reports use a neutral register. And because a wrong rule will be spread persuasively, Tier 1 pairs must be validated before use in high-stakes reports.

Taylor, Pham, Rivkin and Armor (1998): Harnessing the imagination

American Psychologist, 53(4), 429–439.

B GRADE

Process simulation
Imagining the specific steps needed to reach a goal produced progress toward goals and better coping.

Outcome simulation
Imagining the goal already achieved did not.

What it means for THE BRAIN. Development advice describes specific situations and specific steps. For an Ambition × Self-Regulation Conflict, the advice is "in the next tense client meeting, pause before responding to criticism; agree in advance with a colleague on a signal", not "imagine yourself as a calm, composed leader". Rehearsal of trigger situations is built into development notes for Contextual Switch and Conflict patterns.

SECTION G

Teams as Cognitive Systems

Taylor, Fernandes and Wraight (2021): Complementary Cognition

Cambridge Archaeological Journal, 32(1), 61–77. A new theoretical proposal.

C GRADE

Main proposal. Humans are individually specialized in different but complementary strategies for searching for information, and our species adapts at the group level by combining them. Any adaptive system must balance exploration, searching widely for new possibilities, and exploitation, making the most of what is already known. Rather than each person balancing these equally, individuals differ in which they lean toward, and cooperation between differently specialized people produces adaptations no individual could.

Exploration	Integrators	Exploitation
Openness-led archetypes: Visionary, Creator, Reflective Thinker	coordinate between them	Conscientiousness-led archetypes: Operator, Executor, Standards Keeper

Limits. A recent and largely theoretical proposal. It should be cited as a stimulating theory, not established fact.

What it means for THE BRAIN.

- **A theoretical frame for team composition.** Openness-led archetypes (Visionary, Creator, Reflective Thinker) lean toward exploration. Conscientiousness-led archetypes (Operator, Executor, Standards Keeper) lean toward exploitation. Integrators coordinate between them. A team of only one kind will either drift or stagnate.
- **Interaction at team level,** as proposed in Chapter 11.
- Corporate Brain's goal tensions, such as growth versus profitability, are exploration–exploitation tensions, and the strategy library's pairing of archetypes is a practical application of complementary search.
- Team-level claims remain design principles until platform data can test whether archetype diversity predicts team outcomes.

SECTION H

Analysis of THE BRAIN's Own Materials

This section applies the same scrutiny to the framework's internal materials: the Trait DNA and interaction workbook, the Big Five workbook, the Corporate Brain Master Sheet and the 4,095-pair matrix. The figures below were computed directly from the files.

The trait inventory

The matrix covers 91 traits in 19 domain codes.

INTRA	Intrapersonal	7	STUDY	Study skills	5
INTER	Interpersonal	7	PROF	Professional skills	5
COG	Cognitive abilities	7	ORG	Organizational culture	4
INT	Domain interests	7	COMM	Communication	4
LEAD	Leadership	6	LSTYLE2	Learning styles (set 2)	4
VALUES	Values and ethics	5	LSTYLE	Learning styles (set 1)	3
BIG5	Big Five domains	5	WELL	Wellness	3
DRIVE	Motivation and drive	5	CX	Client and customer	2
THINK	Thinking styles	5	PARENT	Parenting	2
EXEC	Execution skills	5		**Total**	**91**

Table V.9 Traits by domain code.

Observations.

The Big Five is the foundation conceptually but a small part numerically. Only 5 of 91 traits are Big Five domains, although the Big Five workbook provides all 30 facets. The other 86 traits are best understood as characteristic adaptations, skills, abilities and interests that sit above the core (in the sense of McAdams and Pals, 2006). Facet analysis will show their relationship to the core.

Overlapping constructs. Emotional Stability overlaps with Stress Tolerance and Resilience; Conscientiousness with Time Management and Result Orientation; Agreeableness with Accommodating and Empathy. Highly correlated traits make interaction effects hard to separate from shared variance. Trait intercorrelations will be reported, and pairs above r = .70 reviewed for merging.

Context-specific domains. Parenting, Study Skills and Learning Styles serve other Brain Interacts products. For organizational use they are optional modules.

A trait under review. One trait in the current matrix, Kapha, refers to a constitutional type in Ayurveda rather than a construct from personality psychology, and has no Trait DNA profile. It is being reviewed: it will either receive a justified profile with supporting evidence or be removed, which would make the inventory 90 traits and 4,005 pairs.

The Tier 1 pairs

The 58 expert-defined pairs are the heart of the framework.

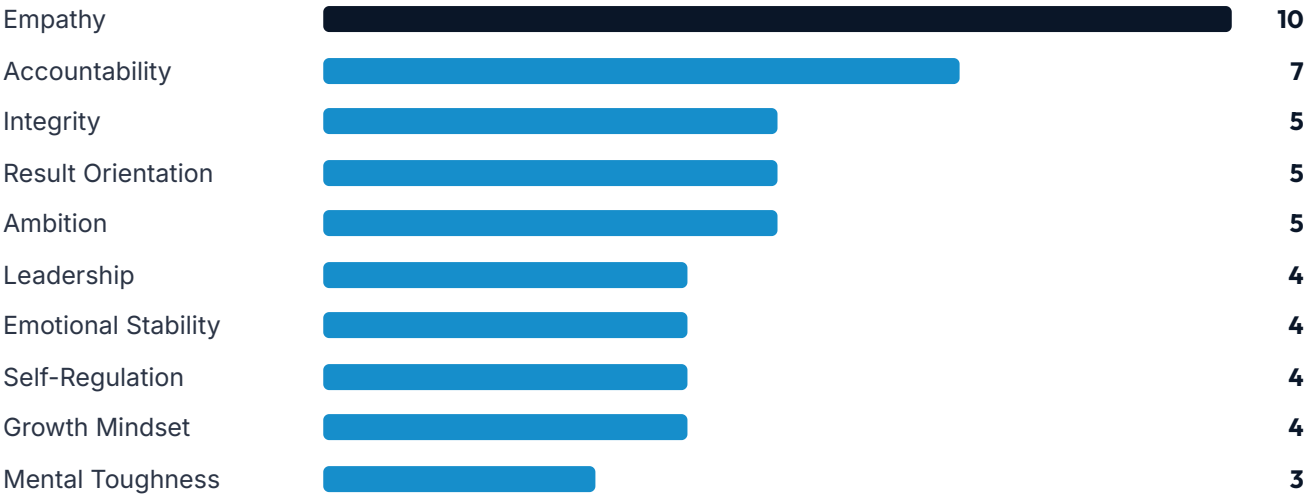


Table V.10 Hub traits in Tier 1 (number of Tier 1 pairs).

Empathy is the framework's central hub. It appears in 10 of 58 pairs and in many of the risk patterns: in Conflict with Mental Toughness, Result Orientation and Assertive Influence; as the missing anchor in the Dominance Risk Multiplier; and in the Contextual Switch with Authoritarian Leadership. In effect, THE BRAIN treats empathy as the trait that keeps drive, power and toughness humane. This is a substantive theoretical position, connected to research on Agreeableness facets such as A6 Tender-Mindedness and A3 Altruism, and to leadership research.

Accountability and Integrity form a second hub. Together they appear in 12 pairs and anchor the two Critical flags, reflecting HEXACO's evidence that honesty-related traits predict ethical conduct beyond the Big Five.

Risk-focused pairs. Fifteen of the 58 pairs (26 per cent) are risk-oriented types: Conflict (5), Risk Multiplier (5), Masking (2), Suppression (2) and Ceiling Effect (1). The remaining 43 are mostly Amplifications. Expanding coverage of risk patterns, which carry the most consequential flags, is a priority for future Tier 1 additions.



Domain crossings. The most common Tier 1 crossings are Drive × Values, Leadership × Values, Interpersonal × Leadership and Interpersonal × Values. Values traits are the most frequent partners, consistent with the framework's view that values channel drive and power.

The Tier 2 rule engine



TIER 2 TYPE	PAIRS	SHARE
Amplification	2,313	96.6%
Compensation	47	2.0%
Conflict	34	1.4%

Table V.11 Tier 2 pairs by interaction type.

The most common pattern label, Cross-Domain Contribution, covers 1,975 pairs (82.5 per cent of Tier 2), with generic insight text.

Problems identified.

Little information

When 97 per cent of rule-generated pairs are labelled Amplification and 83 per cent share one description, Tier 2 adds little beyond saying the traits coexist. All 11 types appear in Tier 1, but only 3 in Tier 2.

Scoring direction is ignored

Some Tier 2 rows pair a risk-scored trait with a strength. Depression × Self-Regulation is labelled Amplification because both are Intrapersonal, but high Depression with high Self-Regulation is closer to Masking.

Conditional logic is missing

The eleven types are defined by score conditions, but Tier 2 labels pairs by domain alone.

REVISIONS UNDER WAY

Most generic Tier 2 pairs are being re-labelled as "no specified interaction; interpret main effects", which is more honest and consistent with Section C. A scoring-direction flag is being added to the Trait DNA, and Tier 2 rules are being made conditional on score patterns, so that the same pair can produce Amplification, Compensation or Conflict depending on scores.

The Tier 3 pairs

The 1,643 Tier 3 pairs are inferred at runtime. The most common crossings involve Interests with Intrapersonal, Cognitive, Interpersonal and Leadership traits, which are reasonable candidates for low priority. Because they are inferred fresh each time, they are the part of the system most exposed to inconsistency across runs and engine versions. Tier 3 inferences are logged, and the most frequently inferred pairs are reviewed for promotion to Tier 1 or Tier 2.

The Big Five workbook

Facet Trait DNA	All 30 NEO facets with high and low behaviour descriptions. This is the framework's most scientifically grounded component.
Behaviour mapping	Each facet is linked to everyday behaviours, situations, high and low signals, a capture method and an inference method. This is a strong research design document.
Facet metrics	Thirty metrics, 27 from project-management software. Two concerns follow. Project tools capture task behaviour well but social and emotional facets poorly, so accuracy will vary by trait. And many metrics reflect job design rather than personality, so metrics must be compared within roles.
Archetypes from high/low splits	Splitting continuous scores discards information and places people just either side of a cut into different archetypes. Two remedies are planned: a "distance from boundary" indicator on every archetype, and empirical profiles derived from platform data for comparison with the 32 theoretical patterns.

The Corporate Brain Master Sheet

6 × 5
goals × functions = 30 cells

30 × 10
cells × core archetypes = 300 strategies

22
family variants mapped with modifiers

26
modifiers

Structure	Six goals × five functions = 30 goal–function cells. Each cell × 10 core archetypes = the 300-entry strategy library. Twenty-two family variants map to core archetypes with modifiers. Twenty-six modifiers adjust management, communication, autonomy and feedback.
Strengths	The goal-first design is distinctive: most assessment platforms start with the person; Corporate Brain starts with what the organization is trying to achieve. The key-tension field for each goal connects naturally to the exploration–exploitation frame in Section G.
Gaps	The 300 strategies are expert-written and untested, and are presented as structured professional guidance with a plan for evaluation. Several modifiers draw on manager observations and 360 feedback, whose data flow needs to be documented.

Summary of internal findings

FINDING	SEVERITY	ACTION
Tier 2 labels 97% of pairs as Amplification	High	Re-label as "no specified interaction"; add conditional rules
Risk-scored traits not reversed in Tier 2	High	Add scoring-direction flag
Kapha lacks profile and scientific basis	High	Justify with evidence or remove
Only 5 of 91 traits are Big Five	Medium	Explain layered model; run facet analysis
Overlapping constructs	Medium	Report intercorrelations; merge where r > .70
Facet metrics rely on project tools	Medium	Validate per trait; compare within role
Archetypes from high/low splits	Medium	Add boundary indicator; compare with empirical profiles
Empathy as central hub	Opportunity	State as a theoretical contribution
Goal-first strategy design	Opportunity	Emphasize as distinctive contribution

Table V.12 Summary of internal findings.

SECTION I

Synthesis

What the evidence supports

The Big Five is the right foundation. Its structure replicates across methods and raters, it predicts job performance with Conscientiousness the most general predictor, and it predicts outcomes from wellbeing to finance to politics.

Facets add detail. The domain-and-facet hierarchy is validated and is already THE BRAIN's structure for the Big Five.

Traits are stable in relative terms, variable in moment-to-moment expression, and changeable over years. Profiles should be read as tendencies, dated and renewed.

Behaviour leaves measurable traces, especially for Openness and Extraversion.

Constrained automated reasoning is the safer design, because a fixed knowledge base limits inconsistency and unexplained conclusions.

Teams need complementary strengths.

What is contested

Whether trait interactions add predictive power

THE BRAIN's distinctive claims have not been tested. This is the main open question of this work.

Behavioural metrics

Whether behavioural metrics measure the intended facets.

Expert-written strategies

Whether expert-written strategies improve outcomes.

Learning styles

Learning styles as predictors, where the evidence is weak.

What the evidence warns about

Privacy

The methods that read personality from behaviour also read protected characteristics.

Persuasive reports

Fluent, narrative reports are trusted beyond their validity.

Short scales

Measuring 91 traits invites short scales, which reduce reliability.

Fairness

Adverse impact must be tested with recognized standards.

Design implications

IMPLICATION	SOURCE(S)	CHAPTER
Present interaction types as interpretive hypotheses; test priority pairs with pre-registered, split-sample designs of 1,000+	Vize and others (2022)	5, 10
Re-label generic Tier 2 pairs; add scoring direction and conditional rules	Section H (Tier 2 rule engine)	6
Review Kapha; report trait intercorrelations	Section H (trait inventory)	6
Add a situation layer	Funder (2006)	5, 11
Write "tends to"; date reports; reassess every 12–24 months	Whole Trait Theory; Holman & Hughes (2021)	7, 10
Show uncertainty bands; flag near-threshold profiles	Rust and others (2020)	6, 7
Separate evidence from interpretation; add "what this does not show"	Reber and others (2004); Green & Appel (2024)	7
Frame development advice as concrete process steps	Taylor and others (1998)	7, 8
Prohibit inference of protected characteristics; audit proxies	Kosinski and others (2013); Gerber and others (2011)	10
Record engine versions; re-run test profiles after updates; audit language by group	Chapter 7 (Traceability)	7, 10
Validate each facet metric within role	Kosinski and others (2013)	8
Apply interaction types at team level	Taylor, Fernandes & Wraight (2021)	9, 11
Test four-fifths rule for every screening trait	Rust and others (2020)	10
Treat learning styles as preferences only	Section D (Sweeney, 2023)	6

Table V.13 Design implications for THE BRAIN and Corporate Brain.

Positioning

WITH CONFIDENCE

The Big Five and its facets are the right foundation for organizational behavioural analytics; profiles must be read as dated tendencies in context; and automated reasoning in assessment must be constrained, traceable and audited.

WITH CARE

That a systematic taxonomy of trait interactions gives practitioners a better way to reason about combinations. Its interpretive value can be argued now. Its predictive value is a hypothesis that Vize and colleagues' results make demanding, and that THE BRAIN's design gives reasons to test.

Saying this plainly makes the work more credible, and more useful.

REFERENCES FOR THE EVIDENCE REVIEW

Aguinis, H., Beaty, J. C., Boik, R. J., & Pierce, C. A. (2005). Effect size and power in assessing moderating effects of categorical variables using multiple regression: A 30-year review. *Journal of Applied Psychology*, 90(1), 94–107.

Barrick, M. R., & Mount, M. K. (1991). *Personnel Psychology*, 44(1), 1–26.

Campbell, Exley, & Doyle (2023). *Financial Services Review*.

Costa, P. T., Jr., & McCrae, R. R. (1995). *Journal of Personality Assessment*, 64(1), 21–50.

Fleeson, W. (2001). Toward a structure- and process-integrated view of personality. *Journal of Personality and Social Psychology*, 80(6), 1011–1027.

Funder, D. C. (2006). *Journal of Research in Personality*, 40(1), 21–34.

Gerber, A. S., Huber, G. A., Doherty, D., & Dowling, C. M. (2011). *Annual Review of Political Science*, 14, 265–287.

Green, M. C., & Appel, M. (2024). *Advances in Experimental Social Psychology*, 70, 1–82.

Holman, D. J., & Hughes, D. J. (2021). *Journal of Occupational and Organizational Psychology*, 94(3), 762–788.

Jayawickreme, E., Zachry, C. E., & Fleeson, W. (2019). Whole Trait Theory. *Personality and Individual Differences*.

Kosinski, M., Stillwell, D., & Graepel, T. (2013). *PNAS*, 110(15), 5802–5805.

McAdams, D. P., & Pals, J. L. (2006). A new Big Five. *American Psychologist*, 61(3), 204–217.

Pashler, H., McDaniel, M., Rohrer, D., & Bjork, R. (2008). Learning styles: Concepts and evidence. *Psychological Science in the Public Interest*, 9(3), 105–119.

Phuthong, T. (2023). *Journal of Human, Earth, and Future*.

Poropat, A. E. (2009). A meta-analysis of the five-factor model of personality and academic performance. *Psychological Bulletin*, 135(2), 322–338.

Reber, R., Schwarz, N., & Winkielman, P. (2004). *Personality and Social Psychology Review*, 8(4), 364–382.

Rust, J., Kosinski, M., & Stillwell, D. (2020). *Modern psychometrics* (4th ed.). Routledge.

Sherman, R. A., & Pashler, H. (2019). Powerful situations: Rare and small.

Sweeney (2023). Doctoral dissertation, St. John's University.

Taylor, H., Fernandes, B., & Wraight, S. (2021). *Cambridge Archaeological Journal*, 32(1), 61–77.

Taylor, S. E., Pham, L. B., Rivkin, I. D., & Armor, D. A. (1998). *American Psychologist*, 53(4), 429–439.

Vize, C. E., Sharpe, B. M., Miller, J. D., Lynam, D. R., & Soto, C. J. (2022). *European Journal of Personality*.

Yu, Chang, & Li (2024). *Education Sciences*, 14, 1050.

REFERENCE MATERIAL

Appendices

A	Trait Inventory by Domain
B	Tier 1 Catalogue Summary
C	Non-negotiable Flags
D	Modifiers and Facet Metrics
E	Research Programme
F	The Complete Interaction Matrix

APPENDIX A

Trait Inventory by Domain

The 91 traits of THE BRAIN, grouped by the domain codes used in the interaction matrix. For each trait, the number of Tier 1 pairs in which it appears is shown on the right. Full high-score and low-score behaviour descriptions and the acquisition and development weights are held in the Trait DNA workbook.

BIG5	Tier 1 pairs
Agreeableness	0
Conscientiousness	1
Emotional Stability	4
Extraversion	0
New Openness	0

COG	
Analytical Thinking	3
Mathematical Intelligence	0
New Clerical Ability	0
New Closure Ability	0
New Numerical Ability	0
New Reasoning Ability	0
New Verbal Ability	0

COMM	
Active Listening	3
Communication	1
Effective Presentation	0
Interpersonal Intellect	0

CX	
Client Servicing	1
Customer Orientation	0

DRIVE	
Ambitious	5
Entrepreneurial Mindset	1
Extrinsic Motivation	0
Initiative	3
Result Orientation	5

EXEC	
Adaptability	1
Ambiguity Management	2
Engagement	0
Managing Relations	0
Time Management	2

INT	
New Agriculture & Env In	0
New Business & Econom In	0
New Creative Art Interest	0
New Home Science In	0
New Physical & Sports In	0
New Science Interest	0
New Social Sci & Hu In	0

INTER	
Accommodating	3
Dominance	2
Empathy	10
Interpersonal Relations	1
Loyalty	1
Prosocial Orientation	1
Respect	1

INTRA	
Depression	2
Hardiness	1
Intrapersonal Intelligence	0
Mental Toughness	3
Resilience	1
Self-Regulation	4
Stress Tolerance	2

LEAD	
Assertive Influence Style	1
Drawing Influence Style	1
Energizing Influence Style	0
Influence	0
Leadership	4
Persuasive Influence	2

LSTYLE	
Authoritarian Leadership	2
Democratic Leadership	2
Laissez Faire Leadership	1

LSTYLE2	
Auditory Learning Style	0
Kinesthetic Intelligence	0
Kinesthetic Learning Style	0
Visual Learning Style	0

ORG	
Foster Change	1
Steadiness	0
Strive for Excellence	1
Work Engagement	3

PARENT	
Active Parenting Style	0
Authoritarian Parenting	0

PROF	
Collaboration	1
Creativity	1
IT Literacy	1
Risk Awareness	1
Technology Readiness	1

STUDY	
New Exam Preaparation	0
New Learning Attitude	1
New Regular Studies	0
New Study Skills	0
New Use of Resources	0

THINK	
Analyst Thinking Style	1
Cognitive Flexibility	2
Growth Mindset	4
Idealist Thinking Style	1
Innovative Thinking	1

VALUES	
Accountability	7
Inclusion	2
Integrity	5
Moral Courage	2
Take Responsibility	0

WELL	
Kapha	0
New Health Habits	1
New Self Care	2

APPENDIX B

Tier 1 Catalogue Summary

The 58 expert-defined pairs are listed in summary form in Table F.2 and in full in the Tier 1 catalogue in Appendix F, with interaction type, pattern name, behavioural insight, risk or watch note and development note for each.

APPENDIX C

Non-negotiable Flags

The twelve flags are listed in Table 6.6 with their priority and their use in hiring and development. Three rules apply to every flag.

Checked first. Flags are checked before any other analysis and placed first in the report.

Never automatic rejection. Flags never produce automatic rejection; they prompt structured human review.

Wellbeing kept out of hiring. Wellbeing Concern and Masking Risk are excluded from hiring entirely and handled only through confidential, consent-based support.

FLAG	TYPICAL TRIGGER PATTERN
Integrity Deficit	Low Integrity, especially alongside high influence or authority traits
Accountability Deficit	Low Accountability, especially in leadership or ownership roles
Reactive Achiever	High Ambition with low Self-Regulation
Power-Blind Authority	High Dominance with low Empathy
Charismatic Risk	High Persuasive Influence with low Integrity
Values Deficit (general)	Low scores across values and ethics traits
Reckless Innovator	High Entrepreneurial Mindset with low Risk Awareness
Motivated but Fragile	High Work Engagement with low Stress Tolerance
Delegation Risk	Patterns indicating difficulty releasing control
Conflict-Avoidant Loyalist	High Loyalty with high Accommodating
Wellbeing Concern	Elevated distress indicators Support only
Masking Risk	Strong functional traits alongside distress indicators Support only

Table C.1 Flag trigger patterns in outline. Exact thresholds are proprietary.

APPENDIX D

Modifiers and Facet Metrics

Corporate Brain defines 26 modifiers. Each specifies a trigger, a data source (questionnaire, workload data, manager observation or 360 feedback), its meaning, and adjustments to management, communication, autonomy and feedback. Selected modifiers appear in Table 8.3. Modifiers fall into four families.

FAMILY	EXAMPLES	TYPICAL ADJUSTMENT
Workload	Overcommitment Risk, Workload Risk	Reduce competing priorities; review commitments
Emotional	Emotional Reactivity, Pressure Sensitivity	Slow escalation; private, specific feedback
Interpersonal	Conflict Risk, Withdrawal	Clarify decision rights; behavioural feedback with examples
Working style	High Autonomy Need, High Structure Need	Agree outcomes or provide explicit plans

Table D.1 Modifier families.

The 30 facet metrics map each NEO facet to a measure drawn from existing work systems: 27 from project management tools and 3 from HRMS, calendar and communication tools. Every metric is treated as a hypothesis until it shows a meaningful within-role correlation with its target facet.

APPENDIX E

Research Programme

Every Research in progress panel in this monograph maps to one study below.

STUDY	PURPOSE	WHAT IS COLLECTED	MINIMUM SAMPLE
Reliability	Internal consistency and test–retest	Item responses; a retest after 2 to 6 weeks	300; 100 for retest
Structure and norms	Factor structure, Indian norms, fairness	Item responses with sector, function, seniority, gender, age	500+
Convergent validity	Agreement with established measures	THE BRAIN plus BFI-2 or IPIP-NEO	200
Criterion and interaction validity	Prediction of outcomes; tests of Tier 1 interactions	Scores plus manager performance ratings	200–300 criterion; 1,000–1,500 interactions
Behavioural metrics	Validation of the 30 facet metrics	Facet scores plus project-tool metrics	150
Narrative engine evaluation	Consistency, fidelity, fairness, expert agreement	Repeated report runs; blind expert ratings	50 profiles, 3 experts
Case studies	Organizational use and outcomes	Client feedback, before and after KPIs, written consent	4–5 clients

Table E.1 Research programme.

The criterion and interaction validity study will pool consented data across client organizations, will be pre-registered, and will use split-sample replication following Vize and colleagues (2022).

APPENDIX F

The Complete Interaction Matrix

The matrix covers all 4,095 pairs of THE BRAIN's 91 traits. This appendix summarizes the matrix by tier, type and domain crossing, and prints the 58 Tier 1 pairs in full. The complete matrix is available to clients and research partners as a supplementary workbook.

Pairs by tier and interaction type

TIER	INTERACTION TYPE	PAIRS
Tier 1	Amplification	**33**
Tier 1	Conflict	**5**
Tier 1	Risk Multiplier	**5**
Tier 1	Compensation	**3**
Tier 1	Masking	**2**
Tier 1	Compounding Development	**2**
Tier 1	Dependency	**2**
Tier 1	Suppression	**2**
Tier 1	Paradox	**2**
Tier 1	Ceiling Effect	**1**
Tier 1	Contextual Switch	**1**
Tier 2	Amplification	**2,313**
Tier 2	Compensation	**47**
Tier 2	Conflict	**34**
Tier 3	Inferred at runtime	**1,643**
Total		**4,095**

Table F.1 Pairs by tier and interaction type.

Tier 1 at a glance

TRAIT 1	TRAIT 2	TYPE	PATTERN
Accountability	Integrity	Amplification	**Principled Accountable**
Accountability	Leadership	Amplification	**Ownership Leader**
Depression	Emotional Stability	Masking	**Masked Vulnerability**
Emotional Stability	Self-Regulation	Amplification	**Emotional Master**
Emotional Stability	Leadership	Amplification	**Steady Leader**
Integrity	Leadership	Amplification	**Principled Leader**
Mental Toughness	Depression	Masking	**Hidden Distress**
Mental Toughness	Self-Regulation	Amplification	**Composed High-Performer**
Accountability	Result Orientation	Amplification	**Ownership Performer**
Accountability	Ambitious	Amplification	**Driven Owner**
Accountability	Initiative	Amplification	**Proactive Owner**
Result Orientation	Integrity	Amplification	**Principled Executor**
Ambitious	Emotional Stability	Compensation	**Resilient Driver**
Ambitious	Self-Regulation	Conflict	**Reactive Achiever**
Work Engagement	Stress Tolerance	Risk Multiplier	**Motivated but Fragile**
Empathy	Leadership	Amplification	**Human-Centred Leader**
Integrity	Moral Courage	Amplification	**Values Warrior**
Resilience	Stress Tolerance	Amplification	**Pressure-Proof**
Persuasive Influence	Integrity	Risk Multiplier	**Charismatic Risk**
Mental Toughness	Empathy	Conflict	**Tough but Blind**
Result Orientation	Ambitious	Amplification	**High-Drive Achiever**
Result Orientation	Empathy	Conflict	**Hard-Driving but Blind**
Ambitious	Growth Mindset	Compounding Development	**Self-Accelerating Achiever**
Strive for Excellence	Work Engagement	Amplification	**Engaged Excellence Driver**
Active Listening	Empathy	Amplification	**Deep Connector**
Active Listening	Communication	Ceiling Effect	**Communication Ceiling**
Initiative	Growth Mindset	Dependency	**Self-Starting Learner**
Persuasive Influence	Empathy	Amplification	**People-First Persuader**
Result Orientation	Time Management	Amplification	**Disciplined Executor**

Table F.2 The 58 Tier 1 pairs (continued overleaf).

TRAIT 1	TRAIT 2	TYPE	PATTERN
New Learning Attitude	Growth Mindset	Amplification	**Development Engine**
Assertive Influence Style	Empathy	Conflict	**Kind but Hesitant**
Empathy	Respect	Amplification	**Trust Architect**
Growth Mindset	Hardiness	Compounding Development	**Antifragile Learner**
Democratic Leadership	Inclusion	Amplification	**Collaborative Leader**
Client Servicing	Empathy	Amplification	**Client Champion**
Time Management	Conscientiousness	Amplification	**Reliable Deliverer**
Dominance	Self-Regulation	Compensation	**Controlled Authority**
New Self Care	Work Engagement	Dependency	**Sustainable High-Engagement**
Adaptability	Ambiguity Management	Amplification	**Change-Ready Professional**
Ambiguity Management	Cognitive Flexibility	Amplification	**VUCA Navigator**
Foster Change	Initiative	Amplification	**Change Champion**
Innovative Thinking	Analytical Thinking	Amplification	**Strategic Innovator**
Interpersonal Relations	Collaboration	Amplification	**Team Amplifier**
Analytical Thinking	Cognitive Flexibility	Amplification	**Adaptive Thinker**
Authoritarian Leadership	Empathy	Contextual Switch	**Situational Commander**
Laissez Faire Leadership	Accountability	Suppression	**Invisible Standards**
Accountability	Accommodating	Suppression	**Harmony Over Standards**
Dominance	Empathy	Risk Multiplier	**Power-Blind Authority**
Active Listening	Drawing Influence Style	Amplification	**Deep Engager**
Inclusion	Prosocial Orientation	Amplification	**Culture Champion**
Analytical Thinking	Creativity	Compensation	**Creative-Analytical Balance**
Authoritarian Leadership	Democratic Leadership	Paradox	**Adaptive Authority**
Accommodating	Moral Courage	Conflict	**Principled People-Pleaser**
Entrepreneurial Mindset	Risk Awareness	Risk Multiplier	**Reckless Innovator**
IT Literacy	Technology Readiness	Amplification	**Digital-Ready Professional**
Analyst Thinking Style	Idealist Thinking Style	Paradox	**Vision-Evidence Tension**
Loyalty	Accommodating	Risk Multiplier	**Conflict-Avoidant Loyalist**
New Health Habits	New Self Care	Amplification	**Sustainable Performer**

Table F.2 The 58 Tier 1 pairs (continued).

Tier 1 catalogue in full

Pairs involving Depression are used only for confidential wellbeing support, never in hiring (Chapter 10).

Principled Accountable

Amplification

Accountability × Integrity · `VALUES × VALUES`

Does what they say and owns mistakes openly. Most foundational combination for organizational trust.

Risk or watch Low — load-bearing structure of any high-trust culture.

Development Develop for culture-bearer and governance roles.

Ownership Leader

Amplification

Accountability × Leadership · `VALUES × LEAD`

Models the accountability they expect from others. Takes full ownership of team outcomes including failures. Trust built through consistency.

Risk or watch Low — highest-value leadership combination.

Development Develop for people-management and senior leadership tracks.

Masked Vulnerability

Masking

Depression × Emotional Stability · `INTRA × BIG5`

Emotional stability conceals depression load from observers and sometimes from the person themselves. Appears fine because functionally stable.

Risk or watch Delayed support. Breakdown severity higher when mask eventually fails.

Development Safe confidential check-in channel. Stability is not a reason to withhold support.

Emotional Master

Amplification

Emotional Stability × Self-Regulation · `BIG5 × INTRA`

Strong baseline stability and strong impulse control together. Remains calm in conflict, composed in crisis, deliberate under ambiguity.

Risk or watch May appear detached. People may interpret composure as indifference.

Development Develop visible warmth cues to complement inner regulation.

Steady Leader

Amplification

Emotional Stability × Leadership · `BIG5 × LEAD`

Leads with calm authority. When pressure peaks this person's composure becomes a stabilizing force. Inspires confidence precisely because they do not react.

Risk or watch May appear too composed — team may perceive low empathy.

Development Strong leadership acquisition signal.

Principled Leader

Amplification

Integrity × Leadership · `VALUES × LEAD`

Leads with values not just authority. Decisions are trusted because they are anchored in something real. The most durable form of organizational influence.

Risk or watch Low — foundational for any senior leadership role.

Development Culture-bearer signal. Develop for governance roles.

Hidden Distress
Masking

Mental Toughness × Depression · `INTRA × INTRA`

Toughness conceals depression load. Performs through distress. Risk is not today's performance — it is eventual collapse after sustained overload.

Risk or watch Delayed intervention. Breakdown severity higher because distress was unmanaged.

Development Create safe confidential support channel. Do not use toughness as reason to withhold care.

Composed High-Performer
Amplification

Mental Toughness × Self-Regulation · `INTRA × INTRA`

Controls impulses and sustains performance under pressure. Neither reactive nor fragile. Most associated with sustained senior performance.

Risk or watch Low — one of the highest-value intrapersonal combinations.

Development Develop for high-pressure or crisis-management roles.

Ownership Performer
Amplification

Accountability × Result Orientation · `VALUES × DRIVE`

Takes full ownership of outcomes including failures while maintaining delivery focus. Does not deflect, does not miss targets without understanding why.

Risk or watch Low — foundational for high-performance culture.

Development Model for team accountability culture.

Driven Owner
Amplification

Accountability × Ambitious · `VALUES × DRIVE`

Combines aggressive goal-pursuit with full ownership of outcomes. Does not just chase targets — takes complete responsibility for whether they land.

Risk or watch Low — highly valuable combination.

Development Strong signal for senior delivery roles.

Proactive Owner
Amplification

Accountability × Initiative · `VALUES × DRIVE`

Takes ownership AND acts without prompting. Identifies what needs to happen, acts, and owns the outcome.

Risk or watch Low — high-value signal for leadership readiness.

Development Develop for senior tracks where self-direction is non-negotiable.

Principled Executor
Amplification

Result Orientation × Integrity · `DRIVE × VALUES`

Delivers results without cutting corners. When pressure mounts to compromise standards for output, this combination holds the line.

Risk or watch Low — highly trusted combination in any role.

Development Strong acquisition signal. Develop for senior leadership.

Resilient Driver
Compensation

Ambitious × Emotional Stability · `DRIVE × BIG5`

High drive cushioned by emotional stability. Pursues goals aggressively without emotional volatility. Pressure activates performance.

Risk or watch May suppress legitimate frustration — emotions go underground.

Development Strong acquisition signal. Monitor for masked stress under sustained demand.

Reactive Achiever
Conflict

Ambitious × Self-Regulation · `DRIVE × INTRA`

Pushes hard and sets aggressive goals but under pressure makes reactive impulsive decisions that damage relationships and judgment. Sprints brilliantly, struggles to sustain.

Risk or watch Burnout, relationship damage, poor decisions under pressure. PRIORITY FLAG.

Development Self-Regulation coaching is single highest ROI intervention for this profile.

Motivated but Fragile

Risk Multiplier

Work Engagement × Stress Tolerance · `ORG × INTRA`

High engagement without stress tolerance creates a burnout formula. Will not disengage before burning out — will simply break.

Risk or watch Burnout. Performance cliff not gradual decline.

Development Resilience building mandatory before role demands increase.

Human-Centred Leader

Amplification

Empathy × Leadership · `INTER × LEAD`

Combines authority with genuine care. Creates followership not compliance. Teams invest in outcomes because they feel seen.

Risk or watch Low — rare and highly valued.

Development Flag for culture-building and transformation roles.

Values Warrior

Amplification

Integrity × Moral Courage · `VALUES × VALUES`

Holds firmly to values AND speaks up when they are violated. Acts on principles under social and political pressure.

Risk or watch May be perceived as inflexible in grey-area cultures.

Development Deploy for governance and ethics-critical roles.

Pressure-Proof

Amplification

Resilience × Stress Tolerance · `INTRA × INTRA`

Bounces back from setbacks AND sustains function during ongoing stress. Not just durable after adversity — performs at high level throughout it.

Risk or watch Low — deploy in most demanding contexts.

Development High-potential signal for senior or turnaround roles.

Charismatic Risk

Risk Multiplier

Persuasive Influence × Integrity · `LEAD × VALUES`

High persuasion without integrity anchor is dangerous. Can move people at scale and make self-serving choices look principled.

Risk or watch Ethical exposure. IMMEDIATE FLAG in any senior role.

Development Ethics interview required. Must be flagged before other findings.

Tough but Blind

Conflict

Mental Toughness × Empathy · `INTRA × INTER`

Pushes through adversity without registering when others are struggling. Does not notice team distress until visible through attrition.

Risk or watch Team disengagement. People feel expendable.

Development Build regular team pulse-check as a deliberate habit.

High-Drive Achiever

Amplification

Result Orientation × Ambitious · `DRIVE × DRIVE`

Does not just want success — builds relentless systems to guarantee outcomes. Both drive and delivery mechanism peak simultaneously.

Risk or watch Burnout if team is not managed. May exhaust others while maintaining own output.

Development Priority acquisition signal. Pair with Self-Regulation check.

Hard-Driving but Blind

Conflict

Result Orientation × Empathy · `DRIVE × INTER`

Delivers outcomes but consistently underweights team experience. Peers experience this person as effective but costly to work with.

Risk or watch Relationship erosion, team attrition, culture damage over time.

Development 360-feedback on impact. Team health as a performance KPI.

Self-Accelerating Achiever

Compounding Development

Ambitious × Growth Mindset · `DRIVE × THINK`

Ambition with growth mindset produces compounding capability. Each goal achieved becomes a development platform for the next.

Risk or watch Low — self-sustaining high-performance profile.

Development Flag for accelerated track. Needs stretch not coaching.

Engaged Excellence Driver

Amplification

Strive for Excellence × Work Engagement · `ORG × ORG`

Highly invested AND refuses average. Brings both energy and standards. Makes quality feel inspiring.

Risk or watch Perfectionism risk. May create unsustainable pace.

Development Develop delegation and iteration mindset.

Deep Connector

Amplification

Active Listening × Empathy · `COMM × INTER`

Understands emotional experience of others AND has listening depth to receive what they share. Trust is both felt and substantiated.

Risk or watch Low — exceptional for people-facing and leadership roles.

Development Deploy in talent development, client success, and culture roles.

Communication Ceiling

Ceiling Effect

Active Listening × Communication · `COMM × COMM`

Communication capability is exceptional but active listening sets a hard ceiling. At senior levels of relationship depth the inability to truly receive caps all communication effectiveness.

Risk or watch Trust erodes at senior levels. People stop sharing because they feel unheard.

Development Active listening is highest-ROI communication investment for this profile.

Self-Starting Learner

Dependency

Initiative × Growth Mindset · `DRIVE × THINK`

Initiative without growth mindset produces restless repetitive action. Growth mindset is the enabler that transforms initiative from busy-ness into compound capability.

Risk or watch Low when both present — powerful combination.

Development Building growth mindset is highest-ROI intervention for high-initiative profiles.

People-First Persuader

Amplification

Persuasive Influence × Empathy · `LEAD × INTER`

Understands audience deeply before influencing. Influence is experienced as help not pressure.

Risk or watch May avoid necessary direct confrontation.

Development Develop assertive communication alongside empathy.

Disciplined Executor

Amplification

Result Orientation × Time Management · `DRIVE × EXEC`

Outcome-focused and structured. Has both the drive to deliver and the planning discipline to ensure it. Reliable, consistent, high-quality.

Risk or watch May over-optimize process at expense of creativity or people.

Development Strong for delivery-critical or operations roles.

Development Engine

Amplification

New Learning Attitude × Growth Mindset · `STUDY × THINK`

Open orientation to learning with genuine belief in development potential. Every experience is converted into growth.

Risk or watch Low — exceptional development profile.

Development Flag for accelerated development and mentoring programs.

Kind but Hesitant
Conflict

Assertive Influence Style × Empathy · `LEAD × INTER`

Deeply attuned to others but over-softens necessary directness. Avoids difficult things because they feel the impact too acutely.

Risk or watch Conflict avoidance. Feedback unclear. Difficult conversations deferred.

Development Develop courageous communication. Build tolerance for short-term discomfort.

Trust Architect
Amplification

Empathy × Respect · `INTER × INTER`

Treats every person with consistent dignity AND understands their emotional experience. Creates psychological safety simply by being present.

Risk or watch Low — high-value for any people-facing role.

Development Exceptional culture signal.

Antifragile Learner
Compounding Development

Growth Mindset × Hardiness · `THINK × INTRA`

Gets stronger under pressure AND treats every challenge as development data. Creates a person who actively benefits from adversity.

Risk or watch Low — exceptional development candidate.

Development Fast-track for stretch assignments.

Collaborative Leader
Amplification

Democratic Leadership × Inclusion · `LSTYLE × VALUES`

Builds genuinely inclusive environments where diverse input shapes decisions. Psychological safety is lived not policy.

Risk or watch Decision speed suffers in fast-moving environments.

Development Excellent for team-building and culture-critical roles.

Client Champion
Amplification

Client Servicing × Empathy · `CX × INTER`

Combines service capability with genuine care for client experience. Anticipates needs rather than reacting to them.

Risk or watch Low — exceptional for client-facing and account roles.

Development Deploy in client success and high-value relationship roles.

Reliable Deliverer
Amplification

Time Management × Conscientiousness · `EXEC × BIG5`

Plans and prioritizes effectively AND has follow-through discipline. Others can depend on commitments being met.

Risk or watch Low — strong acquisition signal for any delivery role.

Development Foundational for roles where reliability is primary performance expectation.

Controlled Authority
Compensation

Dominance × Self-Regulation · `INTER × INTRA`

Natural commanding presence tempered by strong impulse control. Directness is deliberate not reactive. Others experience this as strength not aggression.

Risk or watch Low — strong leadership combination.

Development Develop for management and senior tracks.

Sustainable High-Engagement
Dependency

New Self Care × Work Engagement · `WELL × ORG`

High engagement without self-care is a trajectory not a steady state. Self-care is the enabler that makes engagement sustainable.

Risk or watch Engagement real and valuable but fragile without self-care foundation.

Development Develop self-care practices before role demands increase.

Change-Ready Professional

Amplification

Adaptability × Ambiguity Management · `EXEC × EXEC`

Embraces shifting priorities AND operates effectively without clarity. Performs at peak within change rather than despite it.

Risk or watch Low — flag for transformation or rapidly evolving environments.

Development Exceptional signal for any dynamic or complex role.

VUCA Navigator

Amplification

Ambiguity Management × Cognitive Flexibility · `EXEC × THINK`

Thinks fluidly AND operates at high level without full information or structure. Thrives precisely where others struggle.

Risk or watch Low — exceptional for senior, strategic, or transformation environments.

Development Deploy in high-change or VUCA leadership roles.

Change Champion

Amplification

Foster Change × Initiative · `ORG × DRIVE`

Proactively drives change AND acts without waiting for permission. Does not observe problems — solves them.

Risk or watch May push change faster than organization can absorb.

Development Develop coalition-building and pacing skills.

Strategic Innovator

Amplification

Innovative Thinking × Analytical Thinking · `THINK × COG`

Grounds creative ideas in evidence. Generates solutions that are both genuinely original and practically viable.

Risk or watch Low — exceptional for strategy, product, and consulting.

Development Deploy in complex problem-solving and innovation roles.

Team Amplifier

Amplification

Interpersonal Relations × Collaboration · `INTER × PROF`

Builds strong relationships AND leverages them for collective output. Makes the team better because of how relationships are used.

Risk or watch Low — exceptional for cross-functional or matrix environments.

Development Deploy where coalition-building is required.

Adaptive Thinker

Amplification

Analytical Thinking × Cognitive Flexibility · `COG × THINK`

Applies logical rigor flexibly across changing contexts. Solves novel problems by applying principles adaptively.

Risk or watch Low — strong cognitive profile for senior roles.

Development Strong signal for strategic and advisory roles.

Situational Commander

Contextual Switch

Authoritarian Leadership × Empathy · `LSTYLE × INTER`

Warm and consultative at baseline. Under pressure the authoritarian mode takes over and empathy drops out. Teams experience two completely different leaders.

Risk or watch Team trust erodes when the switch is unpredictable.

Development Name trigger conditions explicitly. Develop conscious mode-switching.

Invisible Standards

Suppression

Laissez Faire Leadership × Accountability · `LSTYLE × VALUES`

Accountability exists in this person's profile but Laissez-Faire style suppresses it. Gives freedom without holding performance to account.

Risk or watch Work falls through cracks. High-autonomy talent thrives, others drift.

Development Build explicit check-in rhythm. Make accountability behaviorally visible.

Harmony Over Standards

Suppression

Accountability × Accommodating · `VALUES × INTER`

Accountability exists but accommodation suppresses it. Has the instinct for standards but harmony need prevents acting on it.

Risk or watch Under-performance tolerance. Difficult conversations perpetually avoided.

Development Create structured contexts where accountability can surface safely.

Power-Blind Authority

Risk Multiplier

Dominance × Empathy · `INTER × INTER`

High dominance amplifies the consequence of low empathy. Not just lacking warmth — a powerful commanding presence that dismisses emotional reality at scale.

Risk or watch Attrition of autonomous talent. Fear-based compliance.

Development 360-feedback mandatory. Team health as performance lever.

Deep Engager

Amplification

Active Listening × Drawing Influence Style · `COMM × LEAD`

Creates genuine space through questioning AND has listening depth to fully receive what comes back. Others feel profoundly heard.

Risk or watch Low — exceptional for coaching, consulting, and facilitation.

Development Deploy in advisory, client-facing, and talent development contexts.

Culture Champion

Amplification

Inclusion × Prosocial Orientation · `VALUES × INTER`

Motivated by collective wellbeing AND actively creates belonging. Culture improves in this person's presence.

Risk or watch Low — exceptional for culture and people leadership.

Development Deploy as culture ambassador and team health steward.

Creative-Analytical Balance

Compensation

Analytical Thinking × Creativity · `COG × PROF`

Imaginative ideation balanced by logical evaluation. Ideas are both novel and viable. Does not generate disconnected from reality.

Risk or watch Low — valuable across functions.

Development Deploy in roles requiring both divergent and convergent thinking.

Adaptive Authority

Paradox

Authoritarian × Democratic Leadership · `LSTYLE × LSTYLE`

Authentically holds both orientations. Directive when execution demands it. Consultative when building or developing. A sophisticated multi-mode leadership profile.

Risk or watch Teams need framework to understand which mode is active.

Development Make switching transparent. Name the modes to the team.

Principled People-Pleaser

Conflict

Accommodating × Moral Courage · `INTER × VALUES`

Has the courage to speak truth under pressure but the need for harmony keeps pulling back. Knows what should be said. Consistently does not say it.

Risk or watch Important truths withheld. Culture issues go unaddressed.

Development Develop structured contexts for practicing courageous communication.

Reckless Innovator

Risk Multiplier

Entrepreneurial Mindset × Risk Awareness · `DRIVE × PROF`

High entrepreneurial energy without risk awareness creates innovator who exposes organization to avoidable harm.

Risk or watch Organizational risk exposure. Resource misallocation.

Development Risk frameworks mandatory before resource authority extended.

Digital-Ready Professional

Amplification

IT Literacy × Technology Readiness · `PROF × PROF`

Comfortable with current technology AND actively embraces emerging tools. Future-proof profile.

Risk or watch Low — strong signal for digital-first organizations.

Development Assign to digital transformation and technology-led projects.

Vision-Evidence Tension

Paradox

Analyst × Idealist Thinking Style · `THINK × THINK`

Authentically holds both evidence-based and visionary orientations. Grounds big ideas in data when it matters, generates vision when logic alone is insufficient.

Risk or watch Without self-awareness can appear contradictory to others.

Development Help person name and make context-switching intentional.

Conflict-Avoidant Loyalist

Risk Multiplier

Loyalty × Accommodating · `INTER × INTER`

Deep loyalty and high accommodation create a profile that protects relationships at the cost of honesty, performance standards, and necessary change.

Risk or watch Groupthink enabler. Feedback suppressor. May enable misconduct by silence.

Development Psychological courage development. Build capacity to challenge without rupturing relationship.

Sustainable Performer

Amplification

New Health Habits × New Self Care · `WELL × WELL`

Maintains both psychological and physical wellbeing practices. Creates biological and emotional foundation for sustained high performance.

Risk or watch Low — wellbeing foundation for any high-demand role.

Development Model for wellbeing culture.

KEY TO INTERACTION TYPES

Amplification · Compensation · Conflict · Suppression · Paradox · Dependency · Ceiling Effect · Masking · Contextual Switch · Compounding Development · Risk Multiplier

Definitions of each type are given in Chapter 5 and summarized in Table 5.1.

Tier 2 patterns by domain crossing

Each Tier 2 pair takes its pattern from the domains it crosses. The table lists every domain crossing that contains Tier 2 pairs, with the number of pairs and the pattern most often assigned. As Part V, Section H, explains, the rule engine is being revised so that these patterns become conditional on score patterns.

Unmarked: Cross-Domain Contribution **DR** Domain Reinforcement **RR** Regulated Relationship **PRF** Personality-Relational Fit **CE** Cognitive Executor **DRT** Drive-Regulation Tension **LE** Leader-Executor **VAP** Values-anchored Performance **RC** Relational Communicator **TE** Thinking Executor

COG × INTRA	49	
COG × INTER	49	
INTER × INTRA	47	RR
INTRA × LEAD	42	
COG × LEAD	42	
INTER × LEAD	39	
INTRA × VALUES	35	
BIG5 × INTER	35	PRF
COG × VALUES	35	
EXEC × INTRA	35	
INTRA × PROF	35	
BIG5 × COG	35	
INTER × THINK	35	
COG × DRIVE	35	
EXEC × INTER	35	
COG × EXEC	35	CE
DRIVE × INTRA	34	DRT
INTRA × THINK	34	
DRIVE × INTER	34	
INTER × PROF	34	
COG × PROF	34	
BIG5 × INTRA	33	
COG × THINK	33	
INTER × VALUES	32	
DRIVE × LEAD	30	
LEAD × THINK	30	
EXEC × LEAD	30	LE
LEAD × PROF	30	
BIG5 × LEAD	29	
COMM × INTRA	28	
INTER × ORG	28	
COG × ORG	28	
COG × COMM	28	
LEAD × VALUES	27	VAP
COMM × INTER	27	RC
INTRA × ORG	27	
BIG5 × VALUES	25	
THINK × VALUES	25	
BIG5 × THINK	25	
EXEC × VALUES	25	
PROF × VALUES	25	
BIG5 × PROF	25	
PROF × THINK	25	
EXEC × PROF	25	
BIG5 × DRIVE	24	
DRIVE × EXEC	24	
LEAD × ORG	24	
BIG5 × EXEC	24	
DRIVE × PROF	24	
EXEC × THINK	24	TE
COMM × LEAD	23	
DRIVE × THINK	23	
DRIVE × VALUES	21	VAP
INTRA × LSTYLE	21	
COG × LSTYLE	21	
COG × COG	21	DR
ORG × VALUES	20	
INTER × LSTYLE	20	
BIG5 × ORG	20	
COMM × VALUES	20	
ORG × THINK	20	
BIG5 × COMM	20	
COMM × DRIVE	20	
EXEC × ORG	20	
ORG × PROF	20	
COMM × THINK	20	
COMM × EXEC	20	
COMM × PROF	20	
DRIVE × ORG	19	
INTRA × INTRA	18	DR
LEAD × LSTYLE	18	
INTER × INTER	18	DR
COMM × ORG	16	
LEAD × LEAD	15	DR
BIG5 × LSTYLE	15	
DRIVE × LSTYLE	15	
LSTYLE × THINK	15	
EXEC × LSTYLE	15	
LSTYLE × PROF	15	
CX × INTRA	14	
COG × CX	14	
LSTYLE × VALUES	13	
CX × INTER	13	
LSTYLE × ORG	12	
COMM × LSTYLE	12	
CX × LEAD	12	
BIG5 × BIG5	10	DR
CX × VALUES	10	
BIG5 × CX	10	
CX × DRIVE	10	
CX × THINK	10	
CX × EXEC	10	
CX × PROF	10	
DRIVE × DRIVE	9	DR
THINK × THINK	9	DR
EXEC × EXEC	9	DR
PROF × PROF	9	DR
VALUES × VALUES	8	DR
CX × ORG	8	
COMM × CX	8	
CX × LSTYLE	6	
ORG × ORG	5	DR
COMM × COMM	5	DR
INTRA × WELL	5	
DRIVE × INT	3	
INT × INTER	3	
DRIVE × STUDY	2	
BIG5 × STUDY	2	
PROF × STUDY	2	
LSTYLE × LSTYLE	2	DR
COG × INT	2	
INT × INTRA	2	
INTER × PARENT	2	
LSTYLE × PARENT	2	
BIG5 × WELL	1	
CX × CX	1	DR
STUDY × THINK	1	
PARENT × VALUES	1	
INTRA × STUDY	1	
EXEC × STUDY	1	
INT × THINK	1	
COMM × PARENT	1	
INT × PROF	1	
ORG × WELL	1	

Table F.4 Tier 2 pairs by domain crossing (pairs; most common pattern).

Tier 3 by domain crossing

Every Tier 3 crossing has the same status: **inferred when relevant** to a profile, from the two traits' DNA profiles.

INT × INTRA	47	COG × WELL	21	COMM × WELL	12
COG × INT	47	INT × LSTYLE	21	LEAD × PARENT	12
INT × INTER	46	INT × WELL	21	INTER × PARENT	12
INT × LEAD	42	INT × INT	21	LSTYLE × LSTYLE2	12
INTER × STUDY	35	ORG × STUDY	20	LSTYLE2 × WELL	12
COG × STUDY	35	COMM × STUDY	20	CX × STUDY	10
INT × VALUES	35	LSTYLE2 × VALUES	20	ORG × WELL	10
BIG5 × INT	35	BIG5 × LSTYLE2	20	BIG5 × PARENT	10
EXEC × INT	35	DRIVE × LSTYLE2	20	DRIVE × PARENT	10
INT × STUDY	35	LSTYLE2 × THINK	20	PARENT × THINK	10
INTRA × STUDY	34	EXEC × LSTYLE2	20	EXEC × PARENT	10
INT × THINK	34	LSTYLE2 × PROF	20	PARENT × PROF	10
INT × PROF	34	LSTYLE2 × STUDY	20	STUDY × STUDY	10
DRIVE × INT	32	LEAD × WELL	18	PARENT × STUDY	10
LEAD × STUDY	30	INTRA × WELL	16	LSTYLE × WELL	9
INT × ORG	28	LSTYLE2 × ORG	16	PARENT × VALUES	9
INTRA × LSTYLE2	28	COMM × LSTYLE2	16	ORG × PARENT	8
COMM × INT	28	LSTYLE × STUDY	15	CX × LSTYLE2	8
INTER × LSTYLE2	28	VALUES × WELL	15	LSTYLE2 × PARENT	8
COG × LSTYLE2	28	DRIVE × WELL	15	COMM × PARENT	7
INT × LSTYLE2	28	THINK × WELL	15	CX × WELL	6
STUDY × VALUES	25	EXEC × WELL	15	PARENT × WELL	6
EXEC × STUDY	24	PROF × WELL	15	LSTYLE2 × LSTYLE2	6
LEAD × LSTYLE2	24	STUDY × WELL	15	LSTYLE × PARENT	4
BIG5 × STUDY	23	BIG5 × WELL	14	CX × PARENT	4
DRIVE × STUDY	23	INTRA × PARENT	14	WELL × WELL	2
STUDY × THINK	23	CX × INT	14	PARENT × PARENT	1
PROF × STUDY	23	COG × PARENT	14		
INTER × WELL	21	INT × PARENT	14		

Table F.5 Tier 3 pairs by domain crossing.

REFERENCE

Glossary

Archetype	A summary label for a pattern of high and low Big Five scores
Consolidation Model	A weighted combination of technical and behavioural scores, set by the client per role
Domain multiplier	A weight applied to a trait pair according to the functional domains it crosses
Facet	One of six narrower traits within each Big Five domain
Flag	A non-negotiable condition checked first, which prompts human review
Interaction type	One of eleven ways in which two traits influence each other's expression
Modifier	A current condition that changes how a manager should respond to an archetype
STEN	A standard score from 1 to 10, mean 5.5, standard deviation 2
Strategy library	300 entries linking goals, functions and archetypes to management actions
Tier 1 / 2 / 3	Expert-defined, rule-generated and engine-inferred trait pairs
Trait DNA	THE BRAIN's structured profile of each trait's behaviour, partners, weights and flags

Brain Interacts

ABOUT BRAIN INTERACTS

Brain Interacts builds behavioural intelligence for organizations. Its platform, Corporate Brain, combines psychometric assessment, technical assessment and work data to help leaders link people decisions to business goals. Research partners and organizations interested in the studies described in this monograph are welcome to contact the team through braininteracts.com.

EMAIL
support@braininteracts.com

WEB
braininteracts.com

© 2026 Brain Interacts. All rights reserved. This monograph may be shared in full for non-commercial purposes with attribution. The THE BRAIN engine instructions, Tier 2 rule weights and item bank are proprietary.